Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns that are not on one straight line will share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

SET 20 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

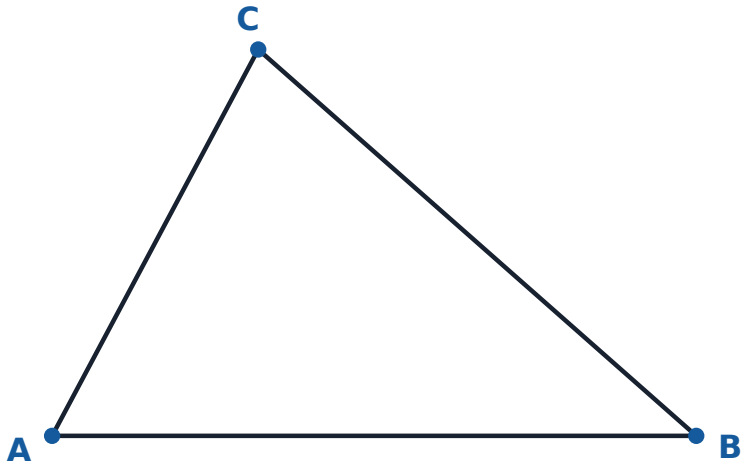
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
$m\angle A + m\angle B + m\angle C = 180°$ in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

In triangle ABC, translate the triangle along ray AB so that A moves to B; C moves to C′ and B moves to D (on ray AB beyond B). Prove that the angles of triangle ABC add to 180°, using the three angles at B.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

In triangle PQR, extend side PQ past Q to S, and draw ray QT parallel to PR (T on the same side of line PS as R). Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

$\angle 1$, $\angle 2$, and $\angle 3$ form a straight angle, with $\angle 1 = \angle 3 = 50°$. Find $\angle 2$.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Triangle ABC has A at the left end of base AB, B at the right end, and C above AB. Line ℓ through C is parallel to AB; D is on ℓ to the left of C and E is on ℓ to the right of C. Which angle is congruent to $\angle CAB$?

A. $\angle ACB$

B. $\angle ECA$

C. $\angle DCA$

D. $\angle ECB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that $m\angle ACD = m\angle A + m\angle B$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

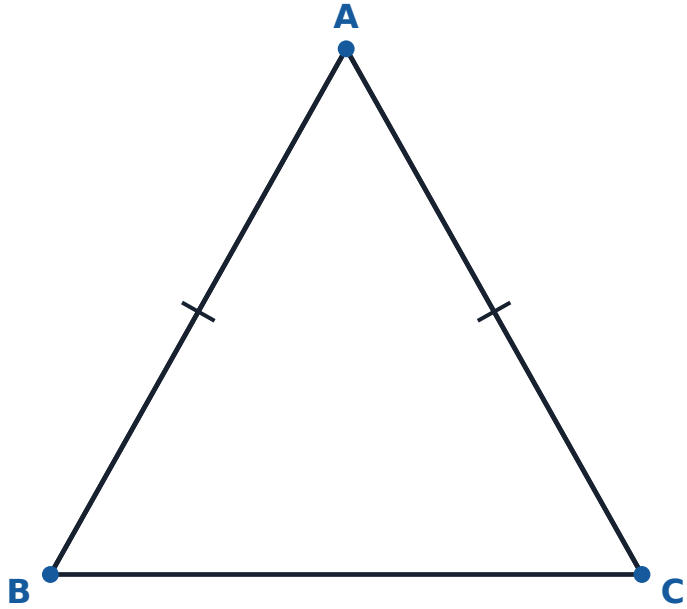
SET 21 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

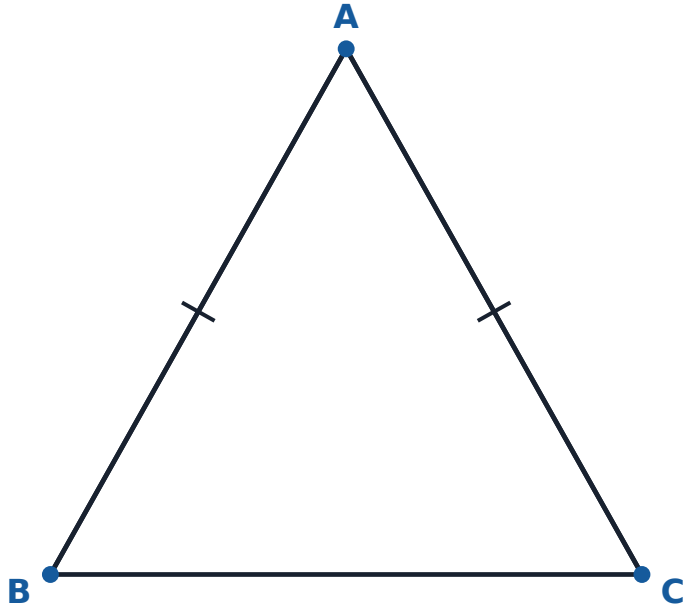
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

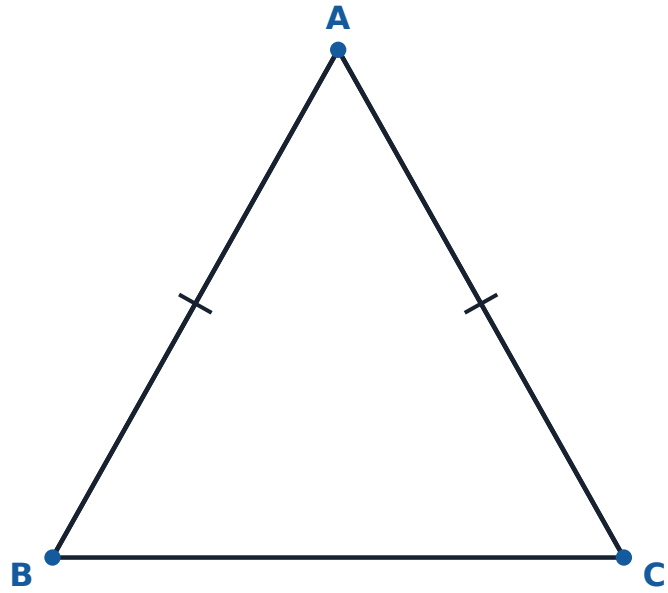
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

Triangle ABC has A(0, h), B(−b, 0), and C(b, 0), with b > 0 and h > 0, so AB = AC. Prove that ∠B ≅ ∠C using the reflection (x, y) → (−x, y).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM. Which criterion shows △ABM ≅ △ACM?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $\triangle ABD \cong \triangle ACD$, then $\angle ABD \cong \angle ACD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of $\angle A$ of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, $\angle B \cong \angle C$. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has $m\angle A = 50°$ and $m\angle B = 65°$. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with $a > 0$ and $c \neq 0$. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

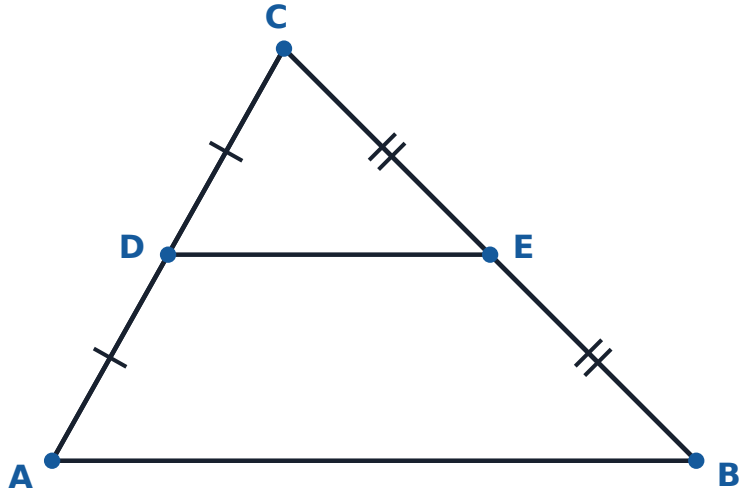
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with $a > 0$ and $c \neq 0$. F is the midpoint of AB and E is the midpoint of AC. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

In triangle ABC, F is the midpoint of AB and E is the midpoint of AC. Extend FE past E to G so that EG = FE. Prove that FE ∥ BC and FE = ½BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

Triangle ABC has A(2p, 2q), B(2r, 2s), and C(2t, 2u), not on one line, with AB not vertical. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from $(-3, 1)$ to $(5, -7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Find the slope of the line through (b, c) and (a + b, c), where $a \neq 0$.

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from $(-2, 5)$ to $(4, 5)$.

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In $\triangle ABC$, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that $EF = DE$, and $\triangle DEC \cong \triangle FEB$. Which fact shows $BF \parallel AC$?

A. $DE = EF$

B. $\angle DCE \cong \angle FBE$ are alternate interior angles

C. $BF = DC$

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5

True or false?

The four triangles formed by drawing all three midsegments of a triangle are similar to it with scale factor ⅓.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q3/5

The midsegment parallel to AB in triangle ABC is 9. A line segment connects the midpoints of that midsegment's triangle the same way. How long is the new segment parallel to AB?

A. 9

B. 18

C. 3

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q4/5

Triangle ABC has vertices A(0, 0), B(8, 0), C(2, 6). Find the vertices of its midsegment triangle and verify the ratio of areas.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q5/5

Prove that the medians of a triangle meet at one point

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints, including with variables	**6.G.A.3**
BACKWARD 2	Use the midsegment theorem	**HSG-CO.C.10**
BACKWARD 3	Find the point a given fraction of the way along a segment	**7.RP.A.2**
BACKWARD 4	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Locate the centroid and use the 2 : 1 ratio	**HSG-GPE.B.6**
FORWARD 2	Prove that the angle bisectors of a triangle meet at one point	**HSG-C.A.3**

SET 23 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2b, 0), and C(2c, 2d), with b ≠ 0 and d ≠ 0. Prove that the point G((2b + 2c)/3, 2d/3) lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q1/5

In triangle ABC, medians BE and CF meet at G. Let H and K be the midpoints of BG and CG. Prove that BG = 2GE. Then explain why the third median also passes through G.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q2/5

Triangle ABC has A(−2a, 0), B(2a, 0), and C(0, 2b), with a > 0 and b > 0. Prove that the three medians meet at (0, 2b/3).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q3/5

Triangle PQR has P(0, 0), Q(6, 0), and R(0, 6). Find the three medians' equations and prove that they meet at one point.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q4/5

Triangle ABC has $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, not on one line. Let $G = ((x_1 + x_2 + x_3)/3, (y_1 + y_2 + y_3)/3)$. Prove that G lies on all three medians.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • MAIN • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 1

Find midpoints, including with variables • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2b, 0) to (2c, 2d).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (0, 6) to (6, 0).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q2/5

A median of a triangle joins a vertex to

A. the midpoint of the opposite side

B. another vertex

C. the foot of the perpendicular from that vertex

D. the midpoint of an adjacent side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q3/5

True or false?

A triangle has exactly three medians.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q4/5

True or false?

A median always bisects the angle at its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 2

Use the midsegment theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, F and E are the midpoints of AB and AC. BC = 14. Find FE.

FE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q1/5

True or false?

The segment joining the midpoints of two sides of a triangle is parallel to the third side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q2/5

In triangle GBC, H and K are midpoints of GB and GC. Which statement is true?

A. HK ⊥ BC

B. HK ∥ GB

C. HK ∥ BC and HK = ½BC

D. HK = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q3/5

A midsegment of a triangle is 11 cm. How long is the side parallel to it?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q4/5

True or false?

A midsegment of a triangle is always half as long as each of the other two sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 3

Find the point a given fraction of the way along a segment • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

On a number line, what point is ⅔ of the way from 0 to 9?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q1/5

On a number line, what point is ⅔ of the way from 3 to 12?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q2/5

A point is ⅔ of the way from A to D. What is the ratio of its distances from A and from D?

A. 2 : 3

B. 3 : 2

C. 1 : 2

D. 2 : 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q3/5

True or false?

The point ⅔ of the way from (0, 0) to (6, 3) is (4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q4/5

True or false?

The point ⅔ of the way from (0, 0) to (3, 6) is (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSG-CO.C.10

BACKWARD 4

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 0) and (3, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q1

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q1/5

Write an equation of the line through (6, 0) and (0, 3).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q2

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q2/5

Which point is on the line $y = -2x + 6$?

A. (1, 2)

B. (3, 3)

C. (0, 2)

D. (2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q3

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q3/5

True or false?

The line through (0, 6) and (3, 0) has slope −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q4

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q4/5

True or false?

The line $y = x$ passes through $(2, 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B4-Q5

Grade 8 • Functions • 8.F.B.4 • Set 23 • BACKWARD 4 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 1

Locate the centroid and use the 2 : 1 ratio • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the point where the medians of the triangle with vertices (1, 2), (7, 4), and (4, 9) meet.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q1/5

Median AD of a triangle is 18 cm long. How far is the centroid from A and from D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

The centroid divides each median so that the part from the vertex is half the part from the midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q3/5

G is the centroid of triangle ABC, and GE = 5 on median BE. Find BE.

A. 20

B. 10

C. 7.5

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q4/5

Explain why the centroid of a triangle is the average of its three vertices.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSG-CO.C.10

FORWARD 2

Prove that the angle bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the bisectors of the three angles of triangle ABC meet at one point that is equidistant from the three sides.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q1/5

Why must the bisectors of $\angle A$ and $\angle B$ of a triangle meet inside the triangle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

A point on the bisector of an angle is equidistant from the two sides of the angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q3/5

The point where the angle bisectors of a triangle meet is the center of the

A. circumscribed circle

B. nine-point circle

C. inscribed circle

D. largest square inside the triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q4/5

True or false?

The angle bisectors and the medians of every triangle meet at the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 23 • FORWARD 2 • Q5/5

Prove that opposite sides of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Alternate interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with ASA	**HSG-CO.B.8**
BACKWARD 3	Know the definition of a parallelogram	**5.G.B.3**
FORWARD 1	Prove that the diagonals of a rhombus are perpendicular	**HSG-CO.C.11**
FORWARD 2	Use opposite sides with coordinates and algebra	**HSG-GPE.B.4**

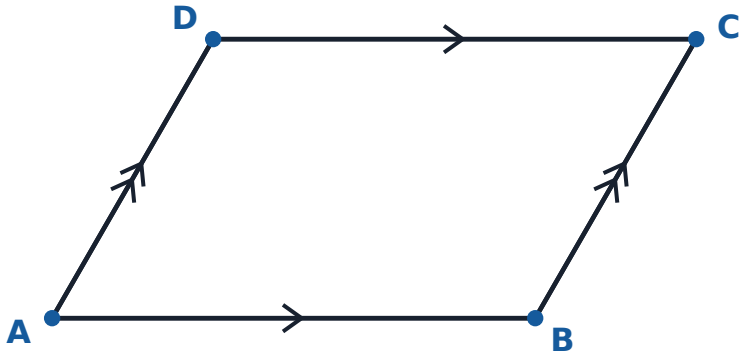
SET 24 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram (AB ∥ DC and AD ∥ BC). Prove that AB = CD and AD = CB. (Draw diagonal AC.)

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q1/5

PQRS is a parallelogram (PQ ∥ SR and PS ∥ QR). Prove that PQ = RS and QR = SP, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q2/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why AB = CD and AD = CB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q3/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c), with a > 0 and c > 0. Prove that AB = DC and AD = BC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q4/5

ABCD and ABEF are parallelograms that share side AB, with C, D, E, F all different points. Prove that DC = FE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • MAIN • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 1

Alternate interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

AB ∥ DC and diagonal AC is drawn. Which angle at C is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q2/5

In parallelogram ABCD with diagonal BD, which angle is congruent to ∠ABD?

A. ∠ABC

B. ∠ADB

C. ∠CBD

D. ∠CDB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q3/5

In parallelogram ABCD with diagonal AC, AD ∥ BC. Which angle is congruent to ∠DAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q4/5

True or false?

Alternate interior angles of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with ASA • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles have two pairs of congruent angles and the congruent sides between them. Which criterion applies?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q1/5

True or false?

A diagonal shared by two triangles can be the included side in ASA.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q2/5

In $\triangle ABC$ and $\triangle CDA$, $\angle BAC \cong \angle DCA$, $\angle BCA \cong \angle DAC$, and AC is shared. Which correspondence is correct?

A. A ↔ D, B ↔ C, C ↔ A

B. A ↔ A, B ↔ D, C ↔ C

C. A ↔ C, B ↔ D, C ↔ A

D. A ↔ C, B ↔ A, C ↔ D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q3/5

After proving $\triangle ABC \cong \triangle CDA$, which side corresponds to $\overline{AB}$?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q4/5

True or false?

ASA requires the side to be between the two angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSG-CO.C.11

BACKWARD 3

Know the definition of a parallelogram • Grade 5 • Geometry • 5.G.B.3

True or false?

A parallelogram is a quadrilateral with both pairs of opposite sides parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q1/5

Which quadrilateral is always a parallelogram?

A. A trapezoid with exactly one pair of parallel sides

B. A kite that is not a rhombus

C. Any quadrilateral with one right angle

D. A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q2/5

Name the two pairs of opposite sides of parallelogram PQRS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q3/5

True or false?

In parallelogram ABCD, sides AB and BC are opposite sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q4/5

Which sides of parallelogram WXYZ are parallel to WX and to XY?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 1

Prove that the diagonals of a rhombus are perpendicular • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rhombus (a parallelogram with all four sides congruent), with diagonals meeting at E. Prove that AC ⊥ BD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q1/5

In rhombus ABCD, diagonals AC = 16 and BD = 12. Find the side length.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q2/5

ABCD is a rhombus. Prove that diagonal AC bisects ∠BAD and ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q3/5

Rhombus ABCD has perimeter 52 and diagonal AC = 24. Find diagonal BD.

BD = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q4/5

In rhombus PQRS, $m\angle PQR = 110°$. Find $m\angle PQS$, where QS is a diagonal, and give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSG-CO.C.11

FORWARD 2

Use opposite sides with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(6, 2), and D(3, 5). Find C, and verify that AB = DC and AD = BC.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q1/5

Explain why the vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c) describe a general parallelogram.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q2/5

In parallelogram WXYZ, $WX = 3x + 4$ and $YZ = 5x - 6$. Find x and WX.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q3/5

For the parallelogram A(0, 0), B(6, 0), C(8, 4), D(2, 4), what is AD?

A. $\sqrt{20} + 2$

B. 4

C. $2\sqrt{5}$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q4/5

Parallelogram ABCD has A(1, 1), B(5, 1), and D(2, 4). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 24 • FORWARD 2 • Q5/5

SET 101 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that opposite angles of a parallelogram are congruent

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Same-side interior angles of parallel lines	**HSG-CO.C.9**
BACKWARD 2	Name opposite and consecutive angles of a quadrilateral	**5.G.B.3**
BACKWARD 3	Prove triangles congruent with ASA and use corresponding angles	**HSG-CO.B.8**
FORWARD 1	Prove angle properties of special parallelograms	**HSG-CO.C.11**
FORWARD 2	Opposite angles and angles of rotation with coordinates	**HSG-CO.A.3**

SET 101 • HSG-CO.C.11

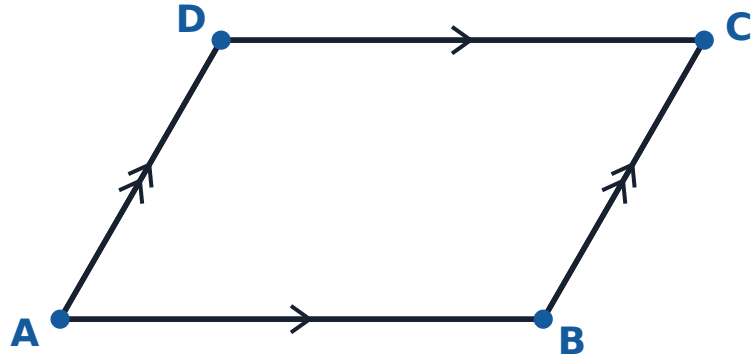
MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram.

Prove that $\angle B \cong \angle D$.

Work space



Conclusion: _____

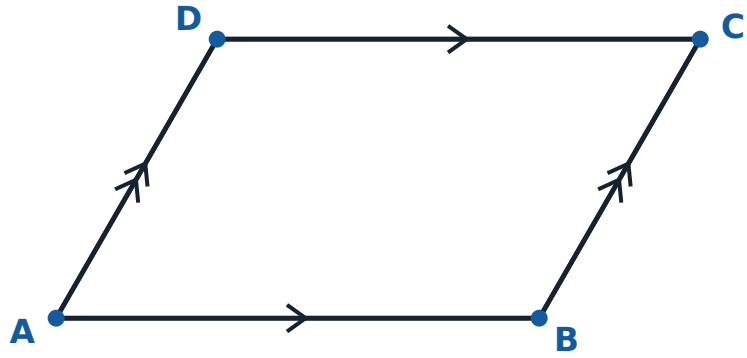
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q1/5

ABCD is a parallelogram. Prove that ∠A ≅ ∠C using same-side interior angles (no diagonal).

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q2/5

PQRS is a parallelogram. Prove that $\angle P \cong \angle R$, using diagonal QS.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q3/5

ABCD is a parallelogram and M is the midpoint of diagonal AC. Explain, using the 180° rotation about M, why ∠DAB ≅ ∠BCD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q4/5

ABCD is a parallelogram with m∠A = x°. Prove that m∠C = x° and m∠B = m∠D = (180 − x)°, giving a reason for each step.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • MAIN • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 1

Same-side interior angles of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Same-side interior angles formed by a transversal of parallel lines are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q1/5

In parallelogram ABCD, side AB is a transversal of AD and BC. What is m∠A + m∠B?

A. It depends on the figure

B. 180°

C. 360°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q2/5

In parallelogram ABCD, m∠A = 65°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal; one same-side interior angle is 72°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q4/5

True or false?

Same-side interior angles of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 101 • BACKWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 2

Name opposite and consecutive angles of a quadrilateral • Grade 5 • Geometry • 5.G.B.3

Name the two pairs of opposite angles of parallelogram WXYZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q1/5

True or false?

In quadrilateral ABCD, $\angle A$ and $\angle B$ are consecutive angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q2/5

Which angle is opposite ∠Q in quadrilateral PQRS?

A. ∠R

B. ∠P

C. ∠Q

D. ∠S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q3/5

Quadrilateral KLMN: name the angles consecutive to ∠K.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q4/5

True or false?

In quadrilateral ABCD, ∠A and ∠C share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 101 • BACKWARD 2 • Q5/5

SET 101 • HSG-CO.C.11

BACKWARD 3

Prove triangles congruent with ASA and use corresponding angles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

$\triangle ABC \cong \triangle CDA$ with $A \leftrightarrow C$, $B \leftrightarrow D$, $C \leftrightarrow A$. Which angle corresponds to $\angle B$?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q1/5

True or false?

After two triangles are proved congruent, their corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q2/5

Which criterion uses two angles and the side between them?

A. ASA

B. SSS

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q3/5

△PQS ≅ △RSQ with P ↔ R. Which angle corresponds to ∠P?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q4/5

True or false?

ASA can use a shared diagonal as its side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 101 • BACKWARD 3 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 1

Prove angle properties of special parallelograms • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram with one right angle has four right angles (so it is a rectangle).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q1/5

In parallelogram ABCD, $m\angle A = (3x + 10)°$ and $m\angle C = (5x - 30)°$. Find x and $m\angle B$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q2/5

Rhombus ABCD has m∠A = 60°. Prove that diagonal BD divides it into two equilateral triangles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q3/5

In parallelogram PQRS, m∠P is twice m∠Q. Find all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q4/5

A parallelogram has an angle of 47°. Which list gives all four angles in order?

A. 47°, 43°, 47°, 43°

B. 47°, 47°, 133°, 133°

C. 47°, 133°, 47°, 133°

D. 47°, 133°, 133°, 47°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 101 • FORWARD 1 • Q5/5

SET 101 • HSG-CO.C.11

FORWARD 2

Opposite angles and angles of rotation with coordinates • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). The 180° rotation about (3, 1.5) maps it onto itself. Which angle does ∠A map onto?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q1/5

Parallelogram ABCD has A(0, 0), B(4, 0), C(6, 3), D(2, 3). Name the image of each vertex under the 180° rotation about (3, 1.5).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q2/5

True or false?

A 180° rotation about the center of a parallelogram maps each angle onto the angle opposite it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q3/5

Which transformation shows that opposite angles of every parallelogram are congruent?

A. A 90° rotation about a vertex

B. A translation along a side

C. A reflection across a diagonal

D. A 180° rotation about the intersection of the diagonals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q4/5

Parallelogram PQRS has P(1, 1), Q(5, 1), R(7, 4), S(3, 4). Give the center and angle of a rotation that maps ∠P onto ∠R.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S101-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 101 • FORWARD 2 • Q5/5

SET 25 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that the diagonals of a parallelogram bisect each other

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Opposite sides of a parallelogram are congruent	**HSG-CO.C.11**
BACKWARD 2	Vertical and alternate interior angles at the crossing of the diago…	**HSG-CO.C.9**
BACKWARD 3	Find the midpoint of a segment and use the definition of bisect	**6.G.A.3**
FORWARD 1	Prove that a 180° rotation carries a parallelogram onto itself	**HSG-CO.A.3**
FORWARD 2	Use diagonal bisection with coordinates and algebra	**HSG-GPE.B.4**

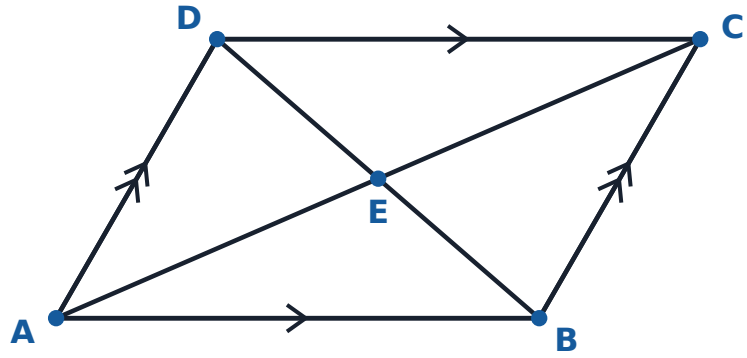
SET 25 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a parallelogram with diagonals AC and BD meeting at E. Prove that AE = CE and BE = DE.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q1/5

PQRS is a parallelogram with diagonals PR and QS meeting at T. Prove that T is the midpoint of both diagonals.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q2/5

A parallelogram has vertices A(0, 0), B(a, 0), C(a + b, c), D(b, c). Prove that its diagonals bisect each other.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q3/5

ABCD is a parallelogram with diagonals meeting at E. Prove that AE = CE and BE = DE using triangles AED and CEB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q4/5

WXYZ is a parallelogram whose diagonals WY and XZ meet at V. Prove that V is the midpoint of WY and of XZ, using the 180° rotation about the midpoint M of WY (show that M is also the midpoint of XZ, so M = V).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • MAIN • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 1

Opposite sides of a parallelogram are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, AB = 9 and BC = 5. Find CD and DA.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q1/5

True or false?

Opposite sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q2/5

In parallelogram WXYZ, WX = 2x + 1 and YZ = x + 6. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q3/5

In parallelogram PQRS, which side is congruent to QR?

A. PQ

B. PR

C. SP

D. RS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q4/5

True or false?

Consecutive sides of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 2

Vertical and alternate interior angles at the crossing of the diagonals • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Diagonals of parallelogram ABCD meet at E. Which angle is vertical to ∠AEB?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q1/5

AB ∥ DC with transversal AC. Which angle is congruent to ∠BAC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q2/5

True or false?

Vertical angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q3/5

AB ∥ DC with transversal BD. Which angle is congruent to ∠ABD?

A. ∠CDB

B. ∠ABC

C. ∠ADB

D. ∠CBD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q4/5

True or false?

$\angle AEB$ and $\angle BEC$ are vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSG-CO.C.11

BACKWARD 3

Find the midpoint of a segment and use the definition of bisect • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (0, 0) to (a + b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q1/5

Find the midpoint of the segment from (a, 0) to (b, c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

A segment bisects another segment if it passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q3/5

Segment AC = 18 is bisected at E. Find AE.

AE = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q4/5

True or false?

If two segments have the same midpoint, each bisects the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 1

Prove that a 180° rotation carries a parallelogram onto itself • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

ABCD is a parallelogram whose diagonals meet at E. Prove that the 180° rotation about E maps ABCD onto itself.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q1/5

Parallelogram ABCD has diagonals meeting at E(2, 3), and A(−1, 1). Find C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q2/5

Quadrilateral ABCD has diagonals that bisect each other at E. Explain why the 180° rotation about E carries ABCD onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q3/5

The 180° rotation about the center of parallelogram PQRS maps side PQ onto

A. side SP

B. side RS

C. side QR

D. diagonal PR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q4/5

A 180° rotation about the intersection of its diagonals carries a parallelogram onto itself. Explain why this implies that opposite angles of a parallelogram are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSG-CO.C.11

FORWARD 2

Use diagonal bisection with coordinates and algebra • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Parallelogram ABCD has A(1, 2), B(7, 2), C(9, 6). Find D and the point where the diagonals meet.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q1/5

In parallelogram ABCD, diagonals meet at E.
AE = 3x − 2 and CE = x + 8. Find AC.

AC = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q2/5

True or false?

The diagonals of every parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q3/5

Three vertices of a parallelogram are (0, 0), (4, 0), (1, 3). Which could be the fourth vertex?

A. (4, 3)

B. (3, 3)

C. (5, 3)

D. (1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q4/5

In parallelogram PQRS, diagonals meet at T. PT = y + 4, TR = 2y − 1, QT = 3y. Find QS.

QS = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove the converses: conditions that make a quadrilateral a parallelogram

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Converses of the parallel-line angle theorems	**HSG-CO.C.9**
BACKWARD 2	Prove triangles congruent with SSS and SAS	**HSG-CO.B.8**
BACKWARD 3	Angle sum of a quadrilateral	**8.G.A.5**
FORWARD 1	Prove that a quadrilateral is a parallelogram using coordinates	**HSG-GPE.B.4**
FORWARD 2	Decide which conditions are enough to guarantee a parallelogram	**HSG-CO.C.11**

SET 26 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In quadrilateral ABCD, AB = CD and AD = CB. Prove that ABCD is a parallelogram. (Draw diagonal AC.)

Work space

Conclusion: _____

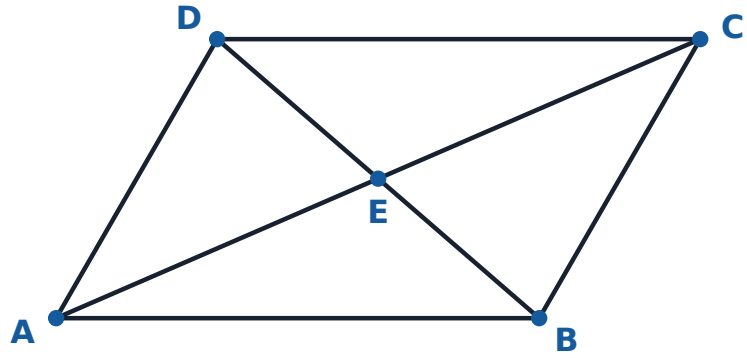
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q1/5

In quadrilateral ABCD, diagonals AC and BD bisect each other at E. Prove that ABCD is a parallelogram.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q2/5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q3/5

In quadrilateral ABCD, $\angle A \cong \angle C$ and $\angle B \cong \angle D$. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q4/5

In quadrilateral ABCD, $\angle A$ and $\angle B$ are supplementary, and $\angle B$ and $\angle C$ are supplementary. Prove that ABCD is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • MAIN • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 1

Converses of the parallel-line angle theorems • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines are cut by a transversal so that alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q1/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q2/5

Which fact proves two lines are parallel?

A. Congruent corresponding angles

B. A linear pair

C. Two acute angles

D. Congruent vertical angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q3/5

Same-side interior angles measure 104° and 76°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q4/5

True or false?

If alternate interior angles are 40° and 50°, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 2

Prove triangles congruent with SSS and SAS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △CDA share AC, and AB = CD, BC = DA. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q1/5

△AEB and △CED: AE = CE, BE = DE, and ∠AEB ≅ ∠CED. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q2/5

True or false?

Vertical angles can serve as the included angle in SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q3/5

After $\triangle ABC \cong \triangle CDA$, which angle is congruent to $\angle BAC$?

A. $\angle BCA$

B. $\angle ABC$

C. $\angle DAC$

D. $\angle DCA$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q4/5

True or false?

SSA proves two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSG-CO.C.11

BACKWARD 3

Angle sum of a quadrilateral • Grade 8 • Geometry • 8.G.A.5

Three angles of a quadrilateral are 80°, 95°, and 110°. Find the fourth.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q1/5

True or false?

The angles of every quadrilateral add to 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q2/5

A quadrilateral has angles x, x, 2x, 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q3/5

Opposite angles of a quadrilateral are 70° and 70°, and another is 110°. Find the fourth.

A. 90°

B. 70°

C. 110°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q4/5

True or false?

A quadrilateral can have four acute angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 1

Prove that a quadrilateral is a parallelogram using coordinates • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(6, 2), C(8, 6), D(3, 5) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 1), R(5, 4), S(0, 3) form a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q2/5

True or false?

If the midpoints of both diagonals of a quadrilateral are the same point, the quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q3/5

Which test alone proves that ABCD is a parallelogram?

A. AC and BD have equal lengths

B. AB $\perp$ BC

C. AB and DC have equal slopes and equal lengths

D. AB and DC have equal lengths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q4/5

A(0, 0), B(5, 0), C(7, 3), D(x, 3) form a parallelogram ABCD. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSG-CO.C.11

FORWARD 2

Decide which conditions are enough to guarantee a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Is a quadrilateral with one pair of opposite sides congruent and the other pair parallel always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q1/5

Is a quadrilateral with both pairs of opposite angles congruent always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with one diagonal that bisects the other must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q3/5

Which condition is NOT enough to prove a quadrilateral is a parallelogram?

A. One pair of opposite sides congruent

B. Both pairs of opposite sides congruent

C. Diagonals bisect each other

D. One pair of opposite sides parallel and congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q4/5

Is a quadrilateral with perpendicular diagonals always a parallelogram? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSG-CO.C.11 • CONGRUENCE: PROVING THEOREMS

Prove that rectangles are parallelograms with congruent diagonals

MAIN	Geometry main question	**HSG-CO.C.11**
BACKWARD 1	Parallelogram properties: opposite sides and consecutive angles	**HSG-CO.C.11**
BACKWARD 2	SAS and SSS with overlapping triangles	**HSG-CO.B.8**
BACKWARD 3	Use the Pythagorean Theorem for diagonals	**8.G.B.7**
FORWARD 1	Decide whether four points form a rectangle	**HSG-GPE.B.4**
FORWARD 2	Prove diagonal properties of rhombuses and squares	**HSG-CO.C.11**

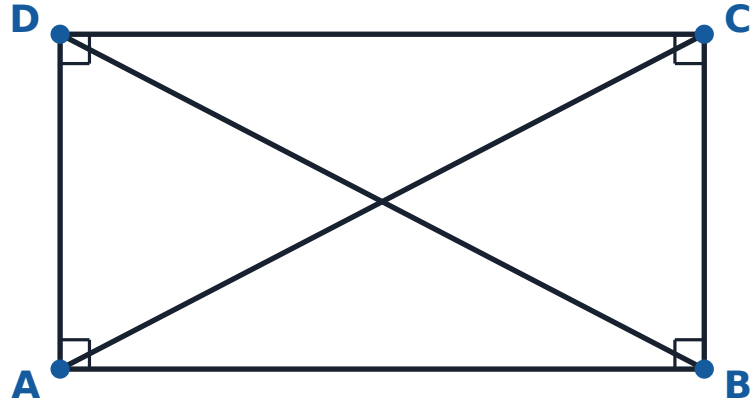
SET 27 • HSG-CO.C.11

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is a rectangle. Prove that AC = BD.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q1/5

ABCD is a parallelogram with AC = BD. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q2/5

Rectangle ABCD has A(0, 0), B(a, 0), C(a, b), D(0, b), with a, b > 0. Prove that its diagonals are congruent.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q3/5

Quadrilateral ABCD has four right angles. Prove that ABCD is a parallelogram (so a rectangle is a parallelogram).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q4/5

The diagonals of quadrilateral ABCD bisect each other and are congruent. Prove that ABCD is a rectangle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • MAIN • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 1

Parallelogram properties: opposite sides and consecutive angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

In parallelogram ABCD, m∠A = 90°. Find the other three angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q1/5

True or false?

Consecutive angles of a parallelogram are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q2/5

In parallelogram PQRS, m∠P = m∠Q. Find m∠P.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q3/5

In parallelogram ABCD, which side is congruent to AD?

A. AC

B. CD

C. AB

D. BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q4/5

True or false?

In every parallelogram, all angles are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

BACKWARD 2

SAS and SSS with overlapping triangles • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

△ABC and △BAD overlap and share AB. Which corresponding angle is at A in △BAD for ∠B in △ABC?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q1/5

True or false?

Overlapping triangles can share a side used in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q2/5

△ABC and △BAD: AB shared, BC = AD, AC = BD. Which criterion?

A. SSS

B. ASA

C. SAS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q3/5

△ABC and △BAD: AB shared, BC = AD, ∠ABC ≅ ∠BAD. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 27 • BACKWARD 2 • Q4/5

True or false?

In $\triangle ABC \cong \triangle BAD$, vertex C corresponds to vertex C.

A. True

B. False

SET 27 • HSG-CO.C.11

BACKWARD 3

Use the Pythagorean Theorem for diagonals • Grade 8 • Geometry • 8.G.B.7

A rectangle is 6 by 8. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q1/5

A rectangle is 5 by 12. Find the length of a diagonal.

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q2/5

True or false?

A rectangle with sides a and b has diagonal $\sqrt{(a^2 + b^2)}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q3/5

A square has side 4. Its diagonal is

A. 8

B. $4\sqrt{2}$

C. $2\sqrt{2}$

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q4/5

True or false?

The two diagonals of a 3-by-7 rectangle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 1

Decide whether four points form a rectangle • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Prove or disprove that A(1, 1), B(5, 3), C(4, 5), D(0, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q1/5

Prove or disprove that P(0, 0), Q(4, 0), R(5, 3), S(1, 3) form a rectangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q2/5

True or false?

A quadrilateral whose diagonals are congruent and bisect each other is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q3/5

Which set of facts proves ABCD is a rectangle?

A. Diagonals bisect each other and are congruent

B. Diagonals are congruent

C. One angle is a right angle

D. Opposite sides are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q4/5

A(0, 0), B(6, 0), and D(0, 4) are three vertices of rectangle ABCD. Find C and verify the diagonals are congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSG-CO.C.11

FORWARD 2

Prove diagonal properties of rhombuses and squares • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Prove that a parallelogram whose diagonals are perpendicular is a rhombus.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q1/5

Which property must a quadrilateral have to be a square, using diagonals only?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q2/5

True or false?

A rectangle with congruent diagonals is a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q3/5

A parallelogram has congruent, perpendicular diagonals. It is a

A. rectangle that is not a square

B. rhombus that is not a square

C. square

D. trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q4/5

A square has diagonal 10. Find its side length (exact) and its area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 27 • FORWARD 2 • Q5/5

Construct a copy of a segment

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Compare and measure lengths	**2.MD.A.1**
BACKWARD 2	A compass draws all points at one distance from its center	**7.G.B.4**
BACKWARD 3	Know rays, segments, and their endpoints	**HSG-CO.A.1**
FORWARD 1	Construct a triangle from three given segments	**HSG-CO.D.12**
FORWARD 2	Decide whether segments can form a triangle, and construct sums	**HSG-CO.D.12**

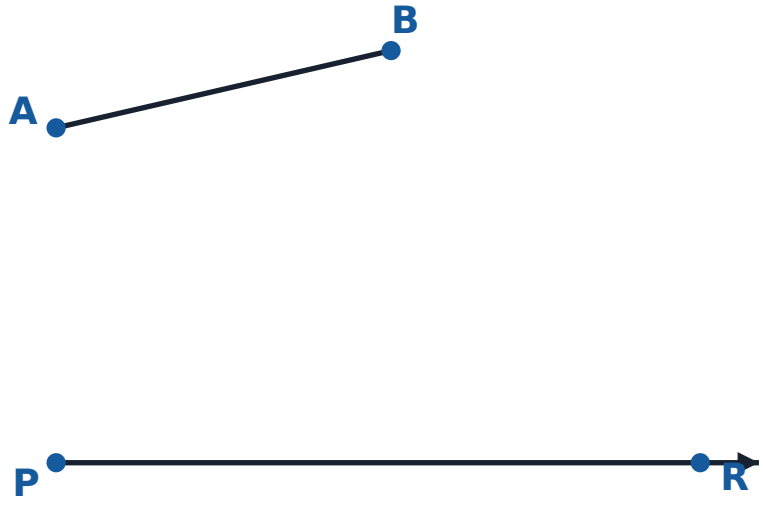
SET 28 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = AB.



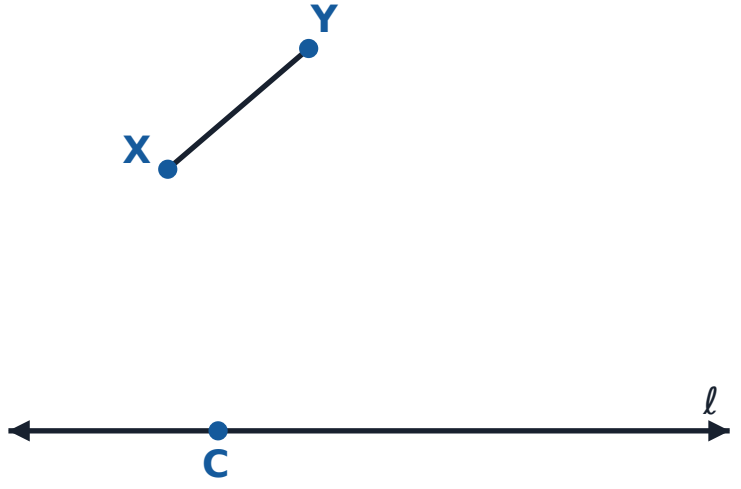
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

On line ℓ, starting at point C, construct a segment CD with CD = XY.



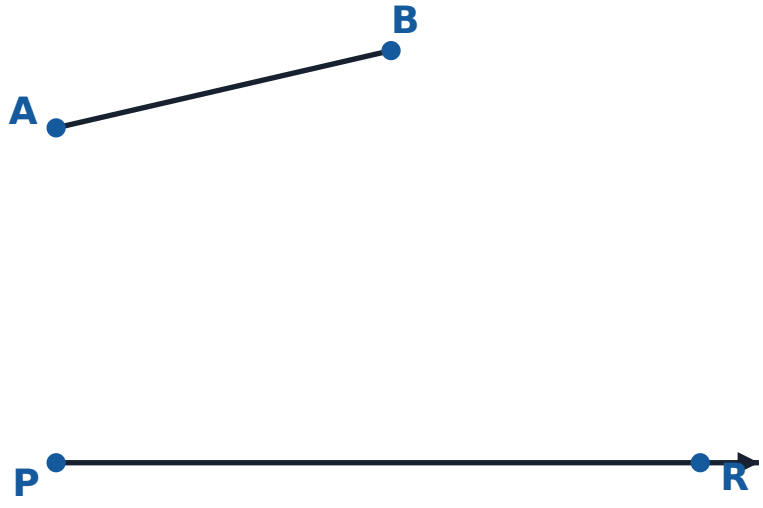
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a point Q on ray PR so that PQ = 2 · AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

On ray XW, construct a point Z so that XZ = AB + CD.





REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct an isosceles triangle with base PQ whose two legs are each congruent to segment AB.





REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • MAIN • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 1

Compare and measure lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment AB is 6 cm and segment CD is 6 cm. Are they congruent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q1/5

True or false?

Two segments are congruent if they have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q2/5

Which tool transfers a length without reading numbers?

A. A compass

B. A calculator

C. A protractor

D. A thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q3/5

A segment 4.5 cm long is copied twice end to end on a line. How long is the result?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A segment drawn longer on the page than another is congruent to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 2

A compass draws all points at one distance from its center • Grade 7 • Geometry • 7.G.B.4

A compass is opened to 5 cm with its point on P. How far from P is every point on the arc it draws?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q1/5

True or false?

Changing the compass opening between steps of a construction keeps the copy accurate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q2/5

An arc centered at P crosses a ray from P at Q. What is PQ?

A. It cannot be known

B. Half the opening

C. Twice the opening

D. The compass opening

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q3/5

True or false?

Every point on a circle is the same distance from the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q4/5

An arc centered at C with radius 3 cm crosses a line through C. How many crossing points are there?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSG-CO.D.12

BACKWARD 3

Know rays, segments, and their endpoints • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Ray PR starts at P and continues through R without end.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q1/5

Q is on ray PR with PQ = 5. Must Q be between P and R? Describe all the cases.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q2/5

Segment AB has endpoints

A. none

B. only B

C. only A

D. A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q3/5

True or false?

A point on ray PR can be on the opposite side of P from R.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q4/5

Name the endpoint of ray PR.

Endpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 28 • BACKWARD 3 • Q5/5

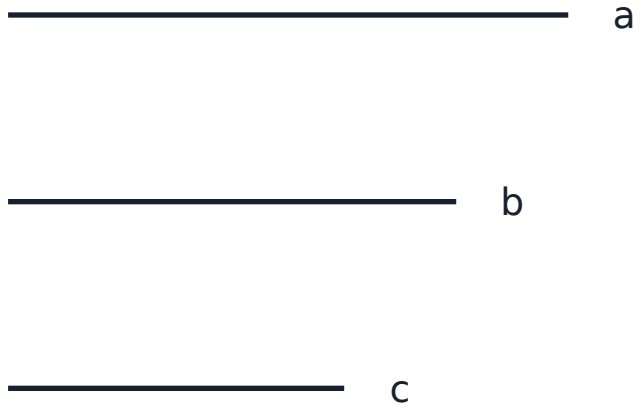
SET 28 • HSG-CO.D.12

FORWARD 1

Construct a triangle from three given segments • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a triangle whose sides are congruent to the three segments a, b, and c.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q1/5

Explain why the two arcs in the triangle construction locate the third vertex.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q2/5

True or false?

Any three segments can be used to construct a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q3/5

To construct a triangle from three given segments, arcs are drawn from the two ends of the base, and they cross at two points, one on each side of the base. The two triangles formed are

A. similar but not congruent

B. different in size

C. not triangles

D. congruent (mirror images)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q4/5

Segments of length 3, 4, and 6 are given. Can a triangle be constructed? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSG-CO.D.12

FORWARD 2

Decide whether segments can form a triangle, and construct sums and differences • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Segments of length 2, 3, and 6 are given. Can a triangle be constructed? Explain what happens to the arcs.

Answer: _____

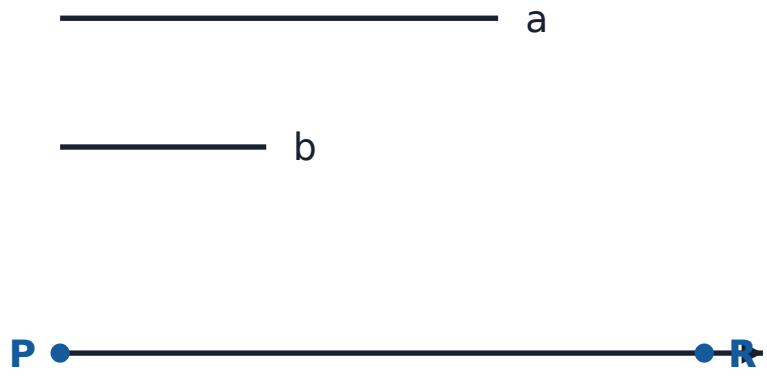
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ID S28-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

On the ray from P, construct a segment of length a − b (a is the longer segment).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q2/5

True or false?

If the arcs in a triangle construction just touch at one point on the base, the "triangle" is flat.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q3/5

Which lengths CANNOT form a triangle?

A. 1, 2, 3

B. 3, 4, 5

C. 4, 4, 7

D. 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q4/5

Two sides of a constructed triangle are 5 and 8. For what third lengths will the arcs meet in two points?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 28 • FORWARD 2 • Q5/5

SET 102 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Make formal constructions with a variety of tools and methods

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Congruent segments have equal lengths	**2.MD.A.1**
BACKWARD 2	A reflection maps a point to its mirror image	**8.G.A.1**
FORWARD 1	Choose and justify a method for a construction	**HSG-CO.D.12**
FORWARD 2	Locate the circumcenter by folding	**HSG-C.A.3**

SET 102 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

You have no compass, only a straightedge and a strip of paper. Describe how to copy segment AB onto line ℓ starting at point C, and explain why the copy is exact.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q1/5

Describe how to construct the perpendicular bisector of segment AB by folding paper, and explain why the crease is the perpendicular bisector.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q2/5

Describe how to bisect an angle drawn on paper by folding, and explain why the crease bisects the angle.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S102-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q3/5

You have a pencil, a pin, and a piece of string, but no compass. Describe how to construct an equilateral triangle on segment AB, and explain why it works.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q4/5

In dynamic geometry software, describe how to construct the line through P parallel to line ℓ so that it stays parallel when P or ℓ is dragged, and explain why a line drawn "by eye" is not a construction.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • MAIN • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 1

Congruent segments have equal lengths • Grade 2 • Measurement & Data • 2.MD.A.1

Segment PQ is 4.5 cm and segment RS is 45 mm. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q1/5

True or false?

A strip of paper can carry a length from one place to another without a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q2/5

Which of these can carry a length from one segment to another without a ruler?

A. A thermometer

B. A calculator

C. A folded paper strip marked at both ends

D. A protractor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q3/5

A string is cut to the length of segment AB. It is laid along line ℓ from C. How long is the segment it marks?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q4/5

True or false?

Two segments with different lengths can be congruent if they are drawn with the same tool.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B1-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 102 • BACKWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

BACKWARD 2

A reflection maps a point to its mirror image • Grade 8 • Geometry • 8.G.A.1

True or false?

Folding paper along a line matches each point with its reflection across the fold.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q1/5

A paper is folded so that point A lands on point B. What is the fold line, in relation to segment AB?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q2/5

A reflection keeps

A. neither

B. lengths and angle measures

C. only lengths

D. only angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q3/5

True or false?

A point on the fold line moves when the paper is folded.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q4/5

Folding makes ray BA land on ray BC. What is true of ∠ABD and ∠DBC, where BD is the crease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 102 • BACKWARD 2 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 1

Choose and justify a method for a construction • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which tool or method would you use to find the midpoint of a segment drawn on tracing paper, without a compass? Explain why it is exact.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q1/5

A reflective device (a transparent mirror) is placed on line ℓ, and point P is seen in it. Explain how to use it to locate the reflection of P across ℓ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q2/5

Explain why measuring a 60° angle with a protractor is not a formal construction, but stepping a compass around a circle is.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q3/5

Describe a way to construct the perpendicular from P to line ℓ by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q4/5

In dynamic geometry software, a student constructs a "square" by placing four points by eye. Describe a drag test that shows it is not a constructed square, and how to construct one properly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 102 • FORWARD 1 • Q5/5

SET 102 • HSG-CO.D.12

FORWARD 2

Locate the circumcenter by folding • High School — Geometry • Geometry: Circles • HSG-C.A.3

Describe how to find the point equidistant from the three vertices of a paper triangle by folding.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q1/5

Why do two creases suffice, without folding the third pair of vertices?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular-bisector creases of a triangle cross can lie outside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q3/5

Folding vertex A onto vertex B of a triangle gives a crease that is

A. the perpendicular bisector of AB

B. the bisector of $\angle C$

C. the median from C

D. parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q4/5

After finding the circumcenter O by folding, how can you draw the circumscribed circle with a pin and string?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S102-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 102 • FORWARD 2 • Q5/5

SET 29 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a copy of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Measure and draw angles with a protractor	**4.MD.C.6**
BACKWARD 2	Copy a segment with a compass	**HSG-CO.D.12**
BACKWARD 3	Arcs and circles in constructions	**7.G.B.4**
FORWARD 1	Justify why the angle-copy construction works	**HSG-CO.B.8**
FORWARD 2	Use angle copying to construct a parallel line	**HSG-CO.D.12**

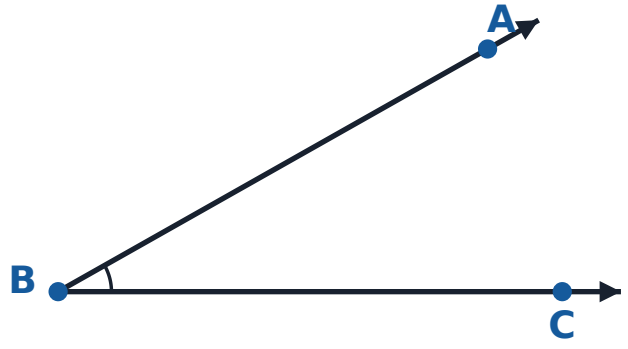
SET 29 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC.





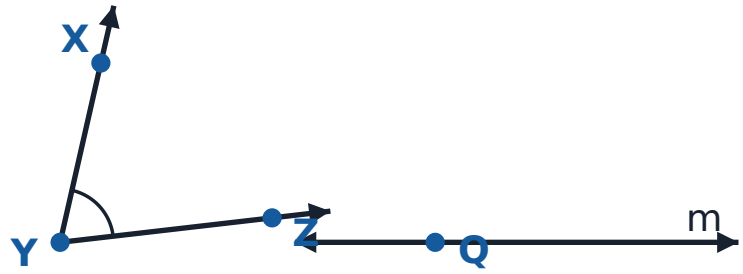
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an angle at point Q on line m congruent to ∠XYZ, opening upward.



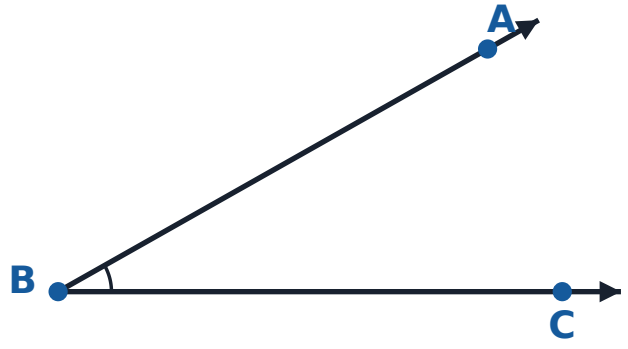
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an angle with vertex P and one side ray PR that is congruent to ∠ABC, opening BELOW ray PR.





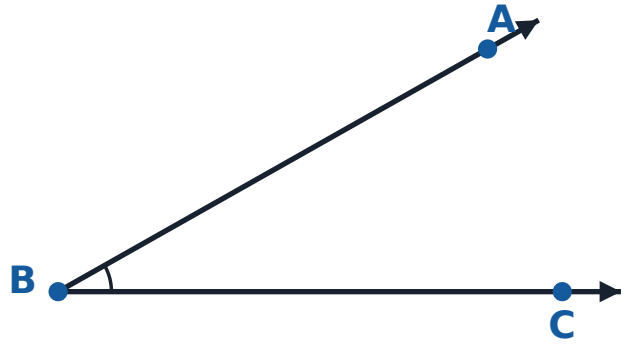
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an angle at P on ray PR whose measure is twice m∠ABC.





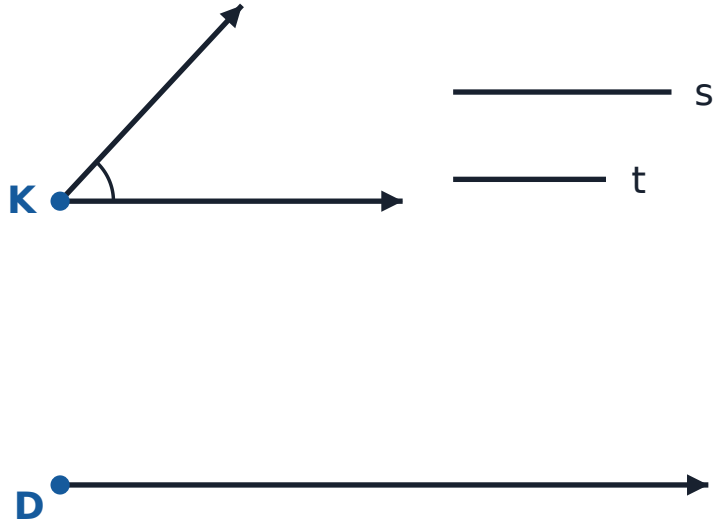
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct triangle DEF with $\angle D \cong \angle K$, DE = s, and DF = t.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • MAIN • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 1

Measure and draw angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor shows a ray at 0° and the other ray at 65°. What is the angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q1/5

True or false?

An angle's measure does not change when its sides are drawn longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q2/5

Which angle is obtuse?

A. 90°

B. 45°

C. 130°

D. 20°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q3/5

Two angles of 35° and 35° are placed side by side. What is the combined angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q4/5

True or false?

Two angles with sides of different lengths cannot be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 2

Copy a segment with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

To copy ∠ABC to vertex P, an arc centered at B crosses BA at D and BC at E, and the same arc centered at P crosses ray PR at F. Which length is set on the compass next, to locate the second side?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q1/5

True or false?

To copy a length, set the compass to the length and do not change it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q2/5

An arc centered at F with radius DE crosses another arc at G. What is FG?

A. Half of DE

B. BD

C. DE

D. PF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q3/5

Arcs centered at B and at P are both drawn with radius 3 cm. How far is F (on the arc at P) from P?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q4/5

True or false?

A segment copy can be checked only by measuring with a ruler.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSG-CO.D.12

BACKWARD 3

Arcs and circles in constructions • Grade 7 • Geometry • 7.G.B.4

True or false?

Two arcs centered at different points can cross at two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q1/5

Points D and E are both on an arc centered at B with radius 4. Find BD and BE.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q2/5

Which describes an arc drawn with a compass?

A. Part of a circle

B. A ray

C. A straight segment

D. A polygon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q3/5

True or false?

An arc centered at P passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q4/5

What does the point of the compass mark in a construction?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSG-CO.D.12

FORWARD 1

Justify why the angle-copy construction works • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In copying ∠ABC to ∠FPG, BD = BE = PF = PG (same arc radius) and DE = FG (same compass setting). Prove that ∠DBE ≅ ∠FPG.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q1/5

∠XYZ is copied to vertex Q, opening downward: an arc of radius r centered at Y meets the sides at U and V, the same arc centered at Q meets the given ray at U′, and an arc of radius UV centered at U′ meets it at V′. Prove that ∠U′QV′ ≅ ∠XYZ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q2/5

∠ABC is copied twice at P: first onto ray PR to give ray PG, then onto ray PG (on the side away from R) to give ray PH. Prove that m∠RPH = 2 · m∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q3/5

In copying ∠ABC to P, a student keeps the arc radius but sets the compass to BD (a radius) instead of the chord DE for the last arc. Which step of the SSS justification fails, and what goes wrong with the copy?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q4/5

Explain why the angle-copy construction works using a rigid motion instead of SSS: there is a rigid motion taking B to P and ray BC onto ray PR.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 29 • FORWARD 1 • Q5/5

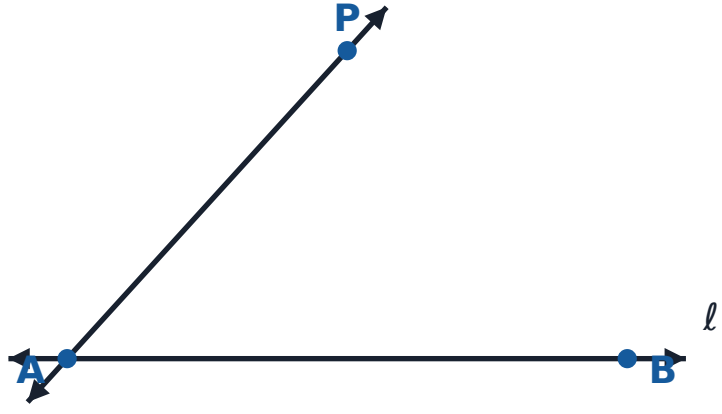
SET 29 • HSG-CO.D.12

FORWARD 2

Use angle copying to construct a parallel line • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an angle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q1/5

Why does copying a corresponding angle at P give a line parallel to ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q2/5

True or false?

Copying the angle so the two angles are congruent same-side interior angles also produces a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q3/5

To construct the line through P parallel to ℓ by copying an angle, a transversal is drawn through P meeting ℓ. Which angle is copied?

A. The angle between ℓ and itself

B. The angle the transversal makes with ℓ

C. A right angle

D. Any 45° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q4/5

Describe a different way to construct a parallel through P using two perpendicular constructions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 29 • FORWARD 2 • Q5/5

SET 30 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the perpendicular bisector of a segment (bisect a segment)

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Points equidistant from A and B lie on the perpendicular bisector	**HSG-CO.C.9**
BACKWARD 2	Know midpoint and perpendicular bisector definitions	**HSG-CO.A.1**
BACKWARD 3	Arcs with one radius from two centers	**7.G.B.4**
FORWARD 1	Justify the perpendicular bisector construction	**HSG-CO.C.9**
FORWARD 2	Construct the circumcenter and circumscribed circle of a triangle	**HSG-C.A.3**

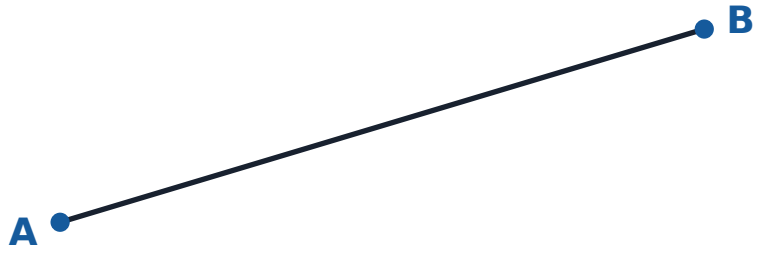
SET 30 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of segment AB. Label the midpoint M.



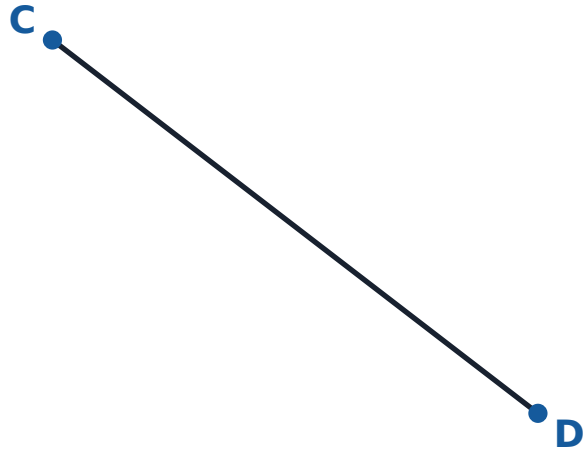
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the midpoint M of segment CD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the point P on line ℓ that is the same distance from A as from B.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular bisector of the short vertical segment UV.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Divide segment AB into four congruent parts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 30 • MAIN • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 1

Points equidistant from A and B lie on the perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

A point equidistant from A and B lies on the perpendicular bisector of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q1/5

Points P and Q are each equidistant from A and B. What line do P and Q determine?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q2/5

In the perpendicular bisector construction for segment AB, the two arc crossings are equidistant from A and B because

A. the crossings are on AB

B. the arcs have the same radius

C. the compass is opened to AB

D. the arcs are drawn above AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q3/5

True or false?

Every point on the perpendicular bisector of AB is equidistant from A and B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q4/5

True or false?

Two points determine more than one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 2

Know midpoint and perpendicular bisector definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB, and AM = 3.5. Find AB.

AB = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector of a segment passes through its midpoint.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q2/5

A line crosses AB at its midpoint at a 60° angle. It is

A. the perpendicular bisector

B. not a bisector

C. parallel to AB

D. a bisector but not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q3/5

Line k is perpendicular to AB and passes through its midpoint. What is k called?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q4/5

True or false?

A segment has many perpendicular bisectors in one plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSG-CO.D.12

BACKWARD 3

Arcs with one radius from two centers • Grade 7 • Geometry • 7.G.B.4

Two arcs of radius 5 are centered at A and B, and AB = 6. Do they cross?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q1/5

True or false?

If the compass opening is less than half of AB, the arcs from A and B do not meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q2/5

Why must the opening be more than half of AB?

A. It does not matter

B. So the midpoint is on the arc

C. So the arcs from A and B cross

D. So the line is longer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q3/5

A point X is on both arcs (radius r) centered at A and B. Find XA and XB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q4/5

True or false?

The two arc crossings are on the same side of AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSG-CO.D.12

FORWARD 1

Justify the perpendicular bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

To construct the perpendicular bisector of AB, arcs of radius r (more than half of AB) centered at A and B meet at P and Q. Prove that line PQ is the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q1/5

Explain why the perpendicular bisector construction for AB would still work with a larger compass opening.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q2/5

True or false?

The construction works only if the arcs are drawn with radius exactly AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q3/5

Which theorem justifies that the arc crossings lie on the perpendicular bisector?

A. Midsegment theorem

B. Triangle angle sum

C. Converse of the perpendicular bisector theorem

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q4/5

Why does folding A onto B produce the perpendicular bisector?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 30 • FORWARD 1 • Q5/5

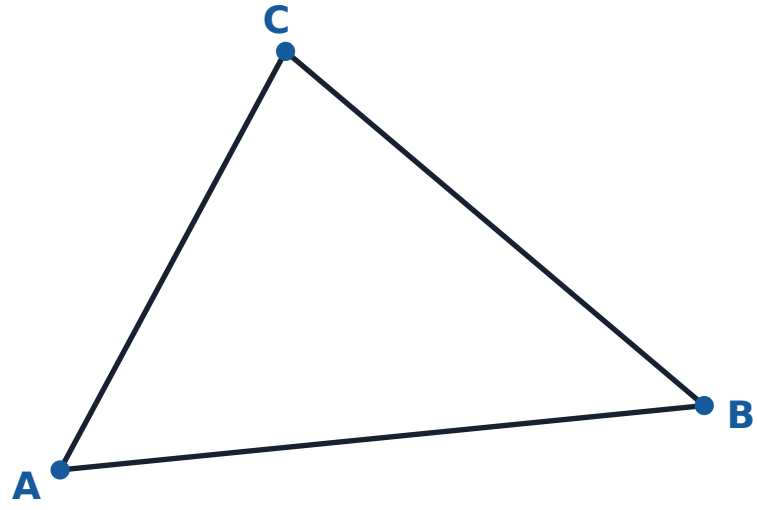
SET 30 • HSG-CO.D.12

FORWARD 2

Construct the circumcenter and circumscribed circle of a triangle • High School — Geometry • Geometry: Circles • HSG-C.A.3

Compass and straightedge only; leave all arcs.

Construct the point O equidistant from A, B, and C, and draw the circle through A, B, and C.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q1/5

Why is it enough to construct only two perpendicular bisectors to find the circumcenter?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q2/5

True or false?

The circumcenter of an obtuse triangle lies on its longest side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q3/5

Where is the circumcenter of a right triangle?

A. Inside, near the centroid

B. At the midpoint of the hypotenuse

C. Outside the triangle

D. At the right-angle vertex

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q4/5

A triangle has A(0, 0), B(8, 0), C(0, 6). Where is its circumcenter, and what is the radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct the bisector of an angle

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of an angle bisector	**HSG-CO.A.1**
BACKWARD 2	Prove triangles congruent with SSS	**HSG-CO.B.8**
BACKWARD 3	Measure angles with a protractor	**4.MD.C.6**
FORWARD 1	Justify the angle bisector construction	**HSG-CO.B.8**
FORWARD 2	Construct special angles and the incenter by bisecting	**HSG-CO.D.12**

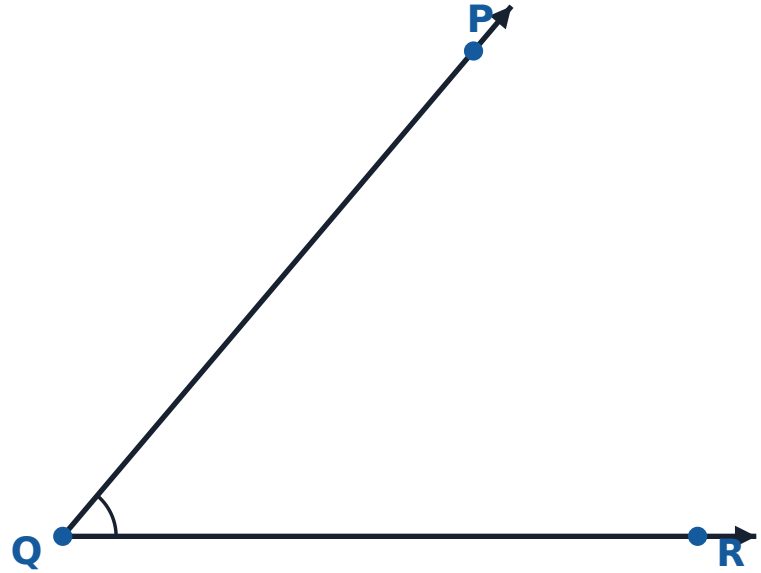
SET 31 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the bisector of $\angle PQR$.



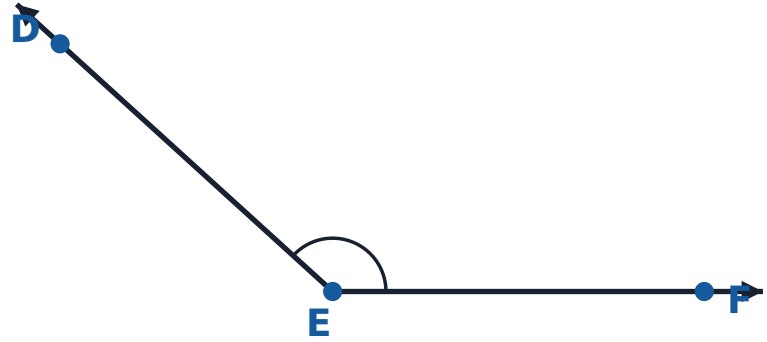
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the bisector of obtuse angle ∠DEF.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a 45° angle with vertex A on line ℓ.



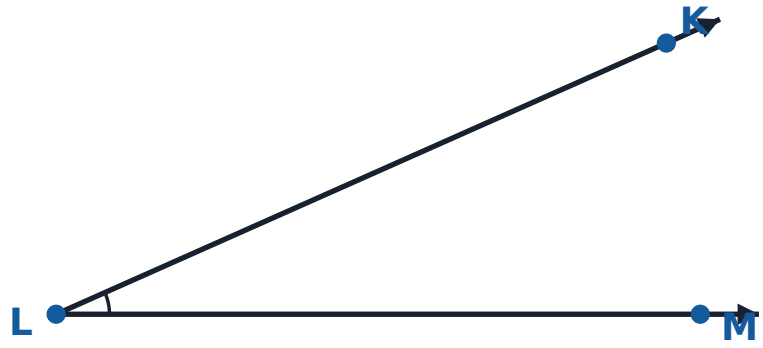
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the bisector of the narrow angle ∠KLM.



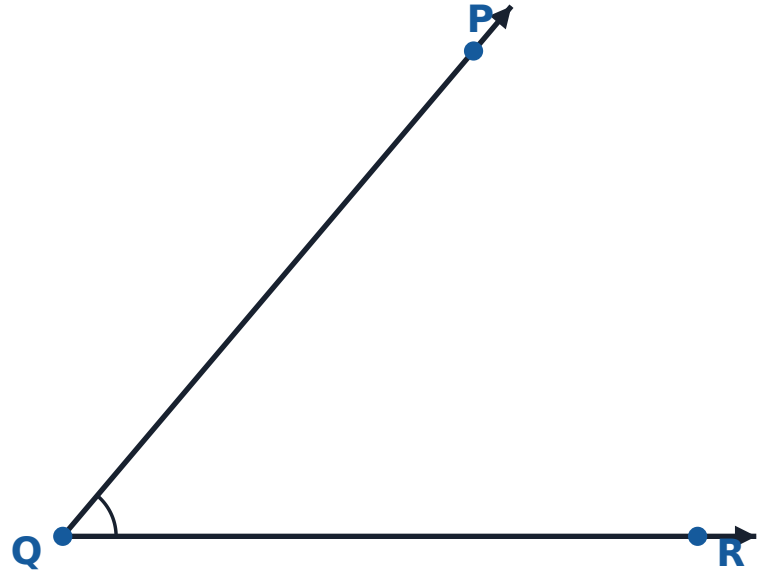
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a ray QS inside ∠PQR so that m∠RQS = ¼ m∠PQR.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • MAIN • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 1

Know the definition of an angle bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray BD bisects ∠ABC, and m∠ABD = 27°. Find m∠ABC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q1/5

True or false?

An angle bisector divides the angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q2/5

Ray BD bisects $\angle ABC$ with $m\angle ABC = 3x + 6$ and $m\angle ABD = 24$. Find x.

A. 6

B. 10

C. 18

D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q3/5

A 90° angle is bisected. What is each part?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q4/5

True or false?

An angle bisector is a segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 2

Prove triangles congruent with SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share a side and have two more pairs of congruent sides. Which criterion?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q1/5

True or false?

SSS guarantees that corresponding angles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q2/5

To bisect ∠ABC, an arc centered at B crosses BA at X and BC at Y, and equal arcs centered at X and Y meet at Z. Which triangles are congruent?

A. △ABC and △XYZ

B. △BXZ and △BYZ

C. △BXY and △XYZ

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q3/5

In the angle-bisector construction for ∠ABC (arc at B meeting the sides at X and Y, equal arcs from X and Y meeting at Z), △BXZ ≅ △BYZ. Which angle is congruent to ∠XBZ?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q4/5

True or false?

SSS requires knowing one angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSG-CO.D.12

BACKWARD 3

Measure angles with a protractor • Grade 4 • Measurement & Data • 4.MD.C.6

A protractor reads 0° on one side and 112° on the other. Half of this angle is

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q1/5

True or false?

Measuring with a protractor is a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q2/5

An angle of 150° is bisected. Each part is

A. 60°

B. 150°

C. 75°

D. 30°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q3/5

After bisecting, a student measures 38° and 38°. What was the original angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q4/5

True or false?

A protractor can check a construction.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.6 • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 1

Justify the angle bisector construction • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In bisecting ∠ABC, an arc centered at B crosses the sides at X and Y, and equal arcs centered at X and Y meet at Z. Prove that ray BZ bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q1/5

In the angle bisector construction, an arc centered at the vertex crosses the sides at X and Y, and then arcs from X and Y cross at Z. Why must the arcs from X and Y have the same radius?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q2/5

True or false?

The bisector construction works for obtuse angles too.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q3/5

Which criterion justifies the angle bisector construction?

A. AAS

B. ASA

C. SSS

D. SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q4/5

Explain why folding one side onto the other gives the bisector.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSG-CO.D.12

FORWARD 2

Construct special angles and the incenter by bisecting • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a 30° angle. (Hint: start from an equilateral triangle.)



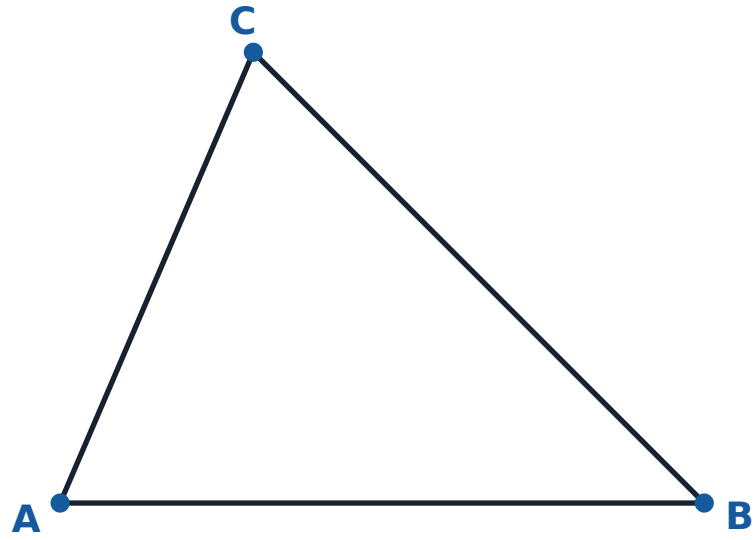
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the point inside triangle ABC that is equidistant from its three sides.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q2/5

True or false?

Bisecting an angle and then bisecting each half divides the angle into three equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q3/5

Starting from 60° and 90° angles, which angle CANNOT be constructed by bisecting angles and adding angles together?

A. 15°

B. 75°

C. 20°

D. 22.5°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q4/5

Explain how to construct a 135° angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a perpendicular to a line through a point on or off the line

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Know the definition of perpendicular lines	**HSG-CO.A.1**
BACKWARD 2	Construct the perpendicular bisector of a segment	**HSG-CO.D.12**
BACKWARD 3	Recognize perpendicular lines in figures	**4.G.A.1**
FORWARD 1	Construct an altitude of a triangle and use it for area	**HSG-CO.D.12**
FORWARD 2	Construct a square on a given segment	**HSG-CO.D.12**

SET 32 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is on ℓ.)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through P perpendicular to line ℓ. (P is not on ℓ.)



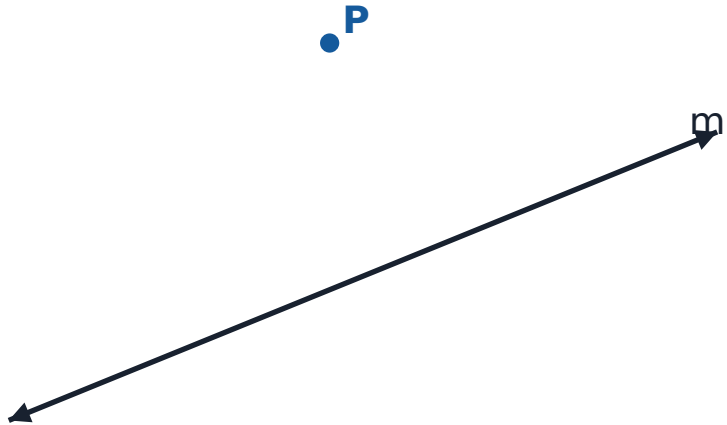
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the perpendicular from point P to the slanted line m.



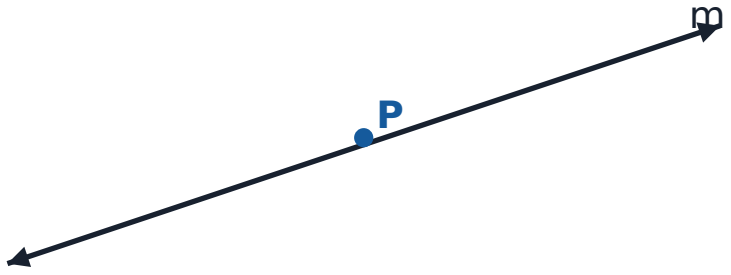
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line perpendicular to the slanted line m at the point P on m.



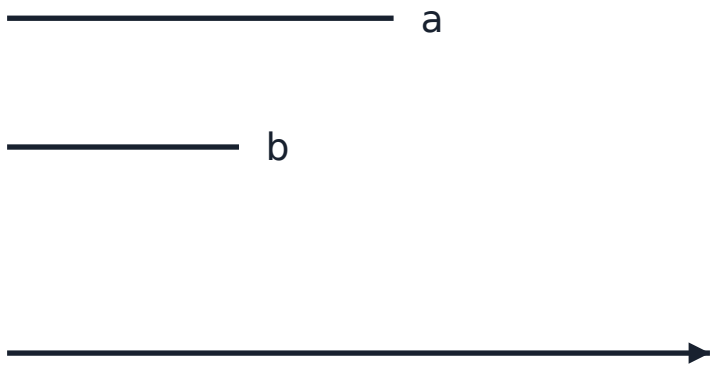
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a right triangle whose legs are congruent to segments a and b.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • MAIN • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 1

Know the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Perpendicular lines meet at a right angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q1/5

Line k $\perp$ line ℓ. What is the measure of each angle they form?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q2/5

Which pair of lines is perpendicular?

A. $y = 3$ and $y = -3$

B. $y = 2x$ and $y = 2x + 1$

C. $y = x$ and $y = 2x$

D. $y = 2x$ and $y = -\frac{1}{2}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q3/5

True or false?

Through a point, there are many lines perpendicular to a given line in a plane.

A. True

B. False

The shortest path from a point to a line follows which kind of line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 2

Construct the perpendicular bisector of a segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

After marking A and B on ℓ, which construction finishes the perpendicular?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q1/5

True or false?

The perpendicular bisector construction needs arcs of equal radius from both endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q2/5

The arc crossings of the perpendicular bisector construction are

A. equidistant from A and B

B. on segment AB

C. on line ℓ

D. at A and B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q3/5

P is on ℓ, and A and B are marked on ℓ with PA = PB. Where does the perpendicular bisector of AB cross ℓ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q4/5

True or false?

The perpendicular bisector of AB passes through A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSG-CO.D.12

BACKWARD 3

Recognize perpendicular lines in figures • Grade 4 • Geometry • 4.G.A.1

True or false?

A square's adjacent sides are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q1/5

Which letter is made of perpendicular segments?

A. V

B. T

C. O

D. S

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q2/5

How many right angles do two perpendicular lines make?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q3/5

True or false?

Two lines that look almost perpendicular must be perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q4/5

Name a pair of perpendicular edges of a rectangular door.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 32 • BACKWARD 3 • Q5/5

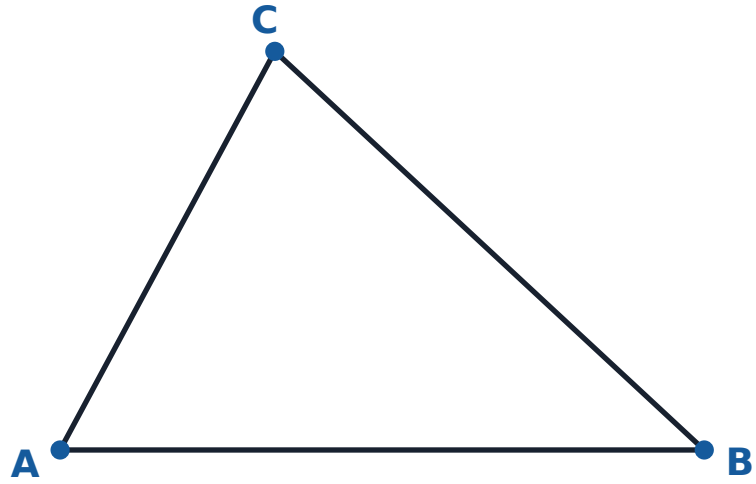
SET 32 • HSG-CO.D.12

FORWARD 1

Construct an altitude of a triangle and use it for area • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the altitude from C to side AB of triangle ABC.



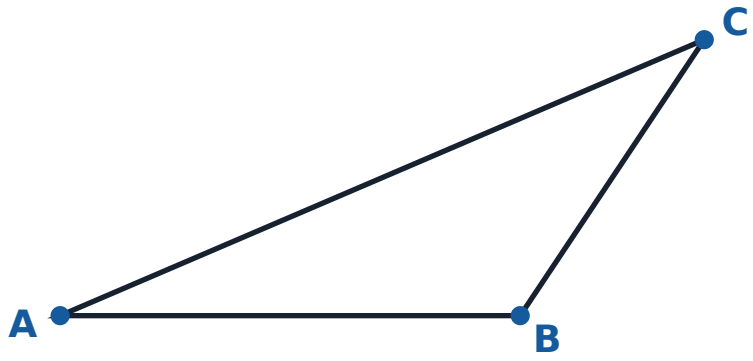
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q1/5

Compass and straightedge only; leave all arcs.

Construct the altitude from C in obtuse triangle ABC (the foot is outside AB). Extend AB as needed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q2/5

A triangle has base 10 and its constructed altitude measures 6. Find its area.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q3/5

True or false?

Every altitude of a triangle lies inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q4/5

In a right triangle, the altitude from the right-angle vertex to the hypotenuse is

A. the hypotenuse

B. inside the triangle

C. outside the triangle

D. a leg

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSG-CO.D.12

FORWARD 2

Construct a square on a given segment • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct a square with side AB, above AB.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q1/5

A square on segment AB is constructed by drawing perpendiculars to AB at A and at B and copying AB onto each, on the same side of line AB, to get D (from A) and C (from B). Explain why ABCD is a square, and why the two copies must be on the same side of AB.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q2/5

True or false?

A quadrilateral with four congruent sides is always a square.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q3/5

To construct a rectangle 2AB long and AB wide, which additional step is needed?

A. Bisect AB

B. Construct a 60° angle

C. Draw a circle through A and B

D. Copy AB twice along the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q4/5

A square is constructed on a segment of length 5. What is the length of its diagonal?

Diagonal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 32 • FORWARD 2 • Q5/5

SET 33 • HSG-CO.D.12 • CONGRUENCE: CONSTRUCTIONS

Construct a line parallel to a given line through a point not on it

MAIN	Geometry main question	**HSG-CO.D.12**
BACKWARD 1	Copy an angle with compass and straightedge	**HSG-CO.D.12**
BACKWARD 2	Converse: congruent corresponding angles give parallel lines	**HSG-CO.C.9**
BACKWARD 3	Recognize parallel lines	**4.G.A.1**
FORWARD 1	Justify the parallel-line constructions	**HSG-CO.C.9**
FORWARD 2	Divide a segment into equal parts with parallel lines	**HSG-SRT.B.4**

SET 33 • HSG-CO.D.12

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ.



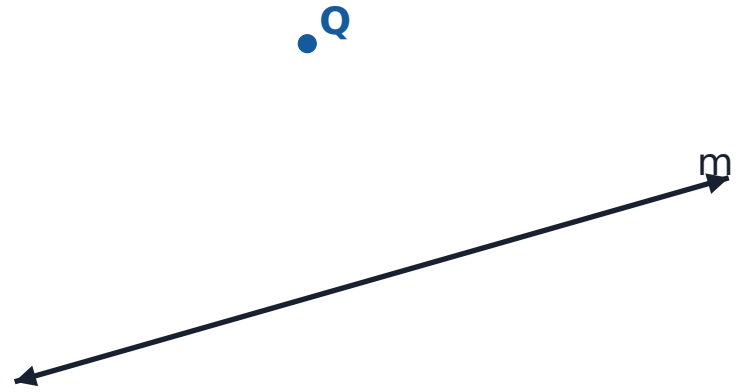
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct the line through Q parallel to the slanted line m.



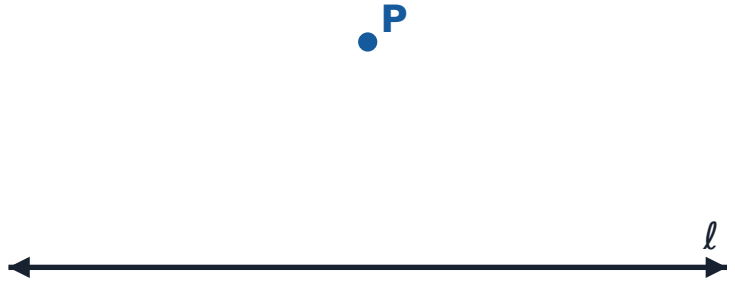
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to ℓ using two perpendiculars.



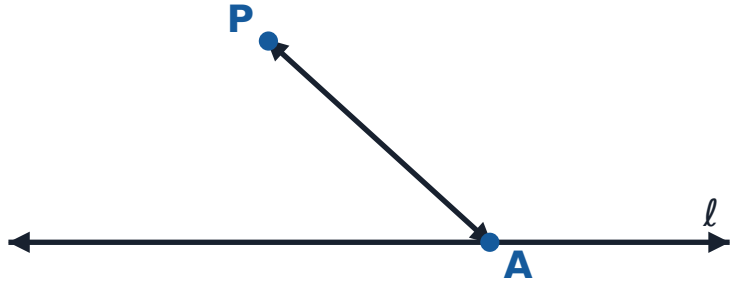
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct the line through P parallel to line ℓ by copying an ALTERNATE INTERIOR angle.



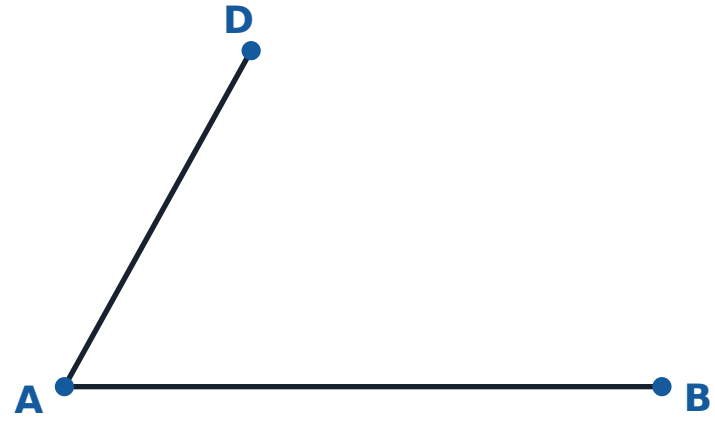
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct parallelogram ABCD with sides AB and AD.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • MAIN • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 1

Copy an angle with compass and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

In copying an angle, which three lengths are transferred?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q1/5

True or false?

An angle copy uses arcs of the same radius at both vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q2/5

Which step completes an angle copy?

A. Draw a circle

B. Measure with a protractor

C. Draw the ray from the new vertex through the last arc crossing

D. Bisect the angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q3/5

Why does the angle copy work?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q4/5

True or false?

An angle copy needs a protractor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 2

Converse: congruent corresponding angles give parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

True or false?

Congruent corresponding angles mean the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q1/5

A transversal makes 62° angles with two lines in corresponding positions. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q2/5

Which also proves the lines parallel?

A. Congruent vertical angles

B. Supplementary linear pairs

C. Congruent alternate interior angles

D. Two acute angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Same-side interior angles of 70° and 70° prove the lines parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q4/5

Corresponding angles are (2x + 10)° and 50°. For which x are the lines parallel?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSG-CO.D.12

BACKWARD 3

Recognize parallel lines • Grade 4 • Geometry • 4.G.A.1

True or false?

The opposite edges of a ruler are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q1/5

Which letter has parallel segments?

A. V

B. A

C. H

D. X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q2/5

Two railroad rails stay the same distance apart. What are they modeling?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q3/5

True or false?

Lines that cross are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q4/5

In a plane, how many lines through a point NOT on a given line are parallel to that line?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 1

Justify the parallel-line constructions • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

To construct a line through P parallel to ℓ, a transversal through P meets ℓ at A, and the angle at A is copied at P in the corresponding position. Prove that the new line is parallel to ℓ.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q1/5

In the two-perpendicular construction, k $\perp \ell$ and n $\perp$ k at P. Prove that n $\parallel \ell$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q2/5

True or false?

Copying the alternate interior angle at P also gives a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q3/5

Which construction gives a line through P parallel to ℓ?

A. Copy, at P, the angle that a transversal through P makes with ℓ, in the corresponding position

B. Bisect the angle at P

C. Draw a circle centered at P

D. Draw the perpendicular bisector of the transversal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q4/5

By the parallel postulate, how many lines could the construction produce?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSG-CO.D.12

FORWARD 2

Divide a segment into equal parts with parallel lines • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

Compass and straightedge only; leave all arcs.

Divide segment AB into three congruent parts.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q1/5

To divide AB into three equal parts, three equal steps are marked on a ray from A, the last mark is joined to B, and parallels to that segment are drawn through the other marks. Why do the parallel lines divide AB into equal parts?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q2/5

True or false?

The same method can divide a segment into 5 equal parts only if its length is a multiple of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q3/5

To divide AB in the ratio 2 : 3, how many equal steps are marked on the auxiliary ray?

A. 2

B. 5

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q4/5

A segment of length 12 is divided into 4 parts this way. How long is each part?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 33 • FORWARD 2 • Q5/5

SET 34 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct an equilateral triangle inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Central angles of a regular polygon	**HSG-CO.A.3**
BACKWARD 2	Copy a length with a compass	**HSG-CO.D.12**
BACKWARD 3	Circle vocabulary: center, radius, chord, diameter	**7.G.B.4**
FORWARD 1	Justify that the inscribed triangle is equilateral	**HSG-C.A.2**
FORWARD 2	Construct a regular dodecagon by bisecting arcs	**HSG-CO.D.13**

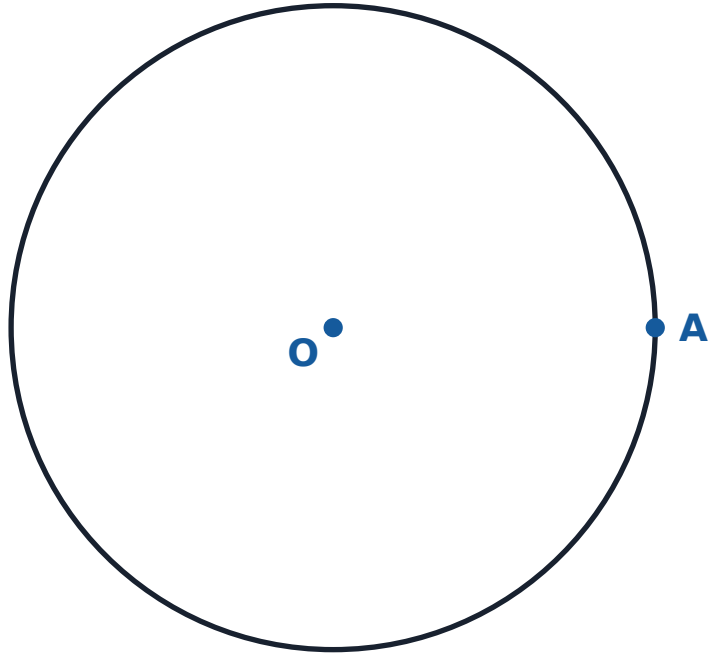
SET 34 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle, with one vertex at A.



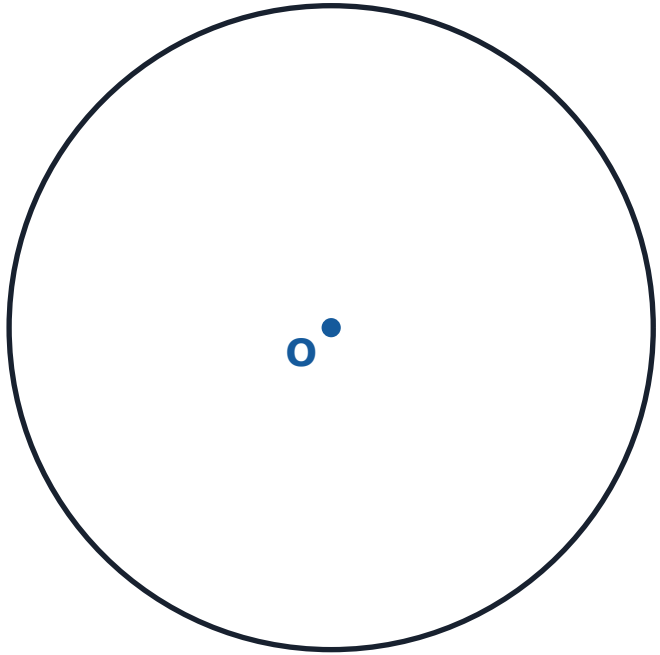
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle (choose any starting vertex).



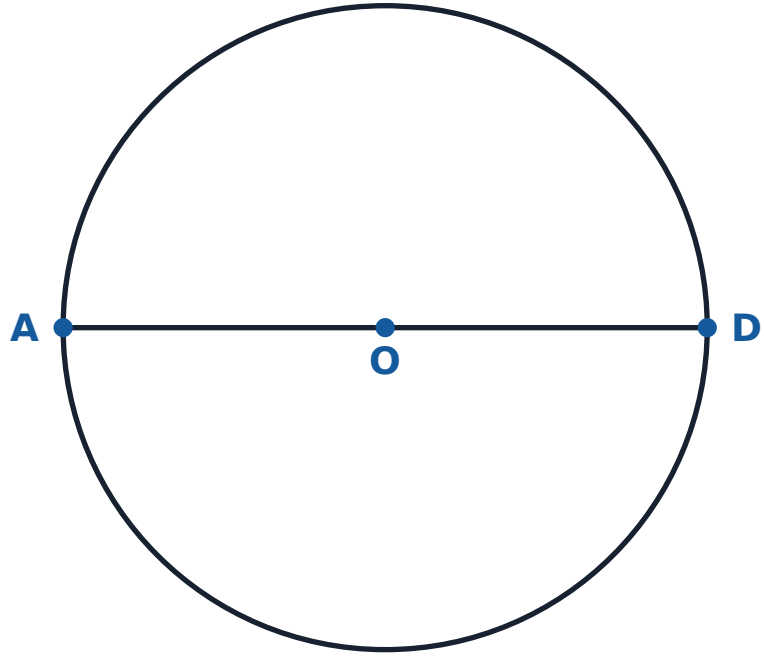
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle using a diameter: draw diameter AD, then an arc centered at D with radius DO.



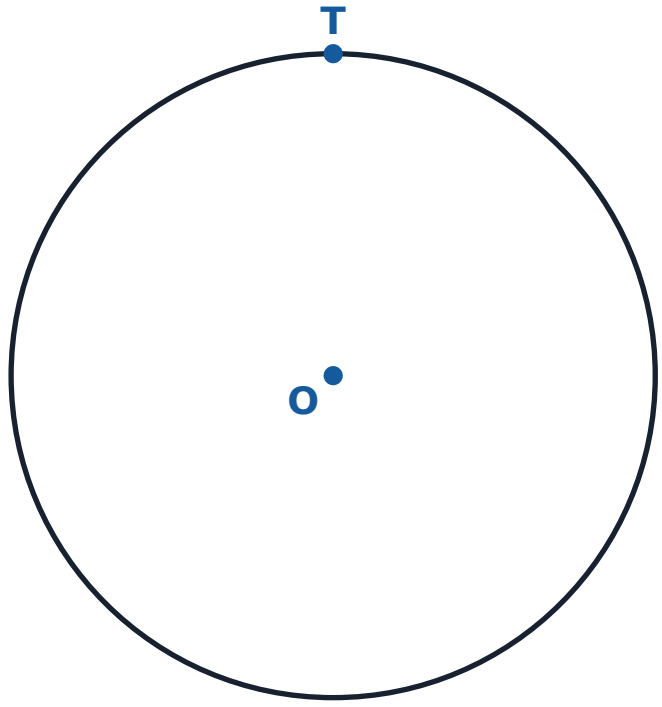
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct an equilateral triangle inscribed in the circle with one vertex at the top point T.



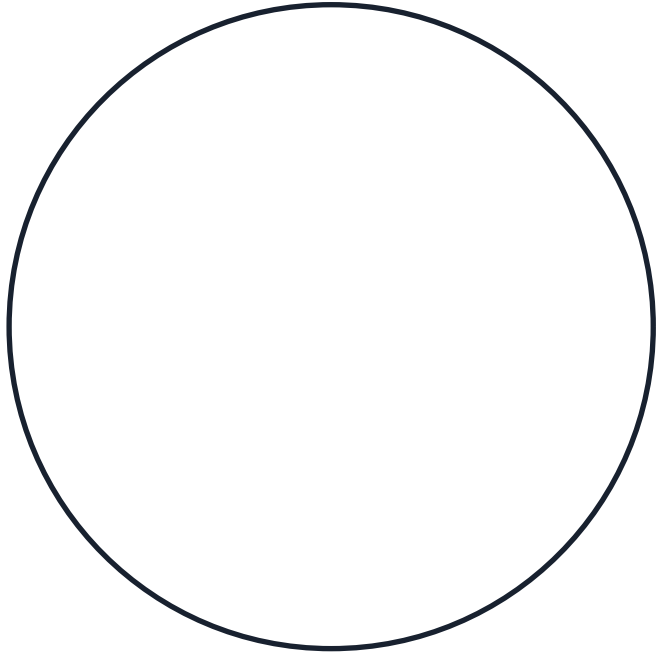
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct an equilateral triangle inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • MAIN • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 1

Central angles of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

What rotation carries an equilateral triangle onto itself, about its center (smallest positive)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q1/5

Into how many equal central angles does a regular hexagon divide 360°?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The vertices of an inscribed equilateral triangle are 120° apart around the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q3/5

Joining every other vertex of a regular hexagon gives

A. a square

B. a rectangle

C. an equilateral triangle

D. a pentagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q4/5

True or false?

The central angles of an inscribed equilateral triangle are 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 2

Copy a length with a compass • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

The compass is set to the radius of a circle. What length does each step around the circle mark?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q1/5

True or false?

Keeping the compass opening fixed is required when stepping around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q2/5

Each radius-length chord and the two radii to its ends form

A. a square

B. an equilateral triangle

C. a right triangle

D. an isosceles triangle with a 90° angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q3/5

A circle has radius 4 cm. How long is each chord marked by stepping the radius?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q4/5

True or false?

Stepping a compass set to the diameter marks six points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSG-CO.D.13

BACKWARD 3

Circle vocabulary: center, radius, chord, diameter • Grade 7 • Geometry • 7.G.B.4

What is a chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q1/5

True or false?

A diameter is a chord through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q2/5

A circle has radius 6. Its diameter is

A. 3

B. 36

C. 6

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q3/5

What does "inscribed in a circle" mean for a polygon?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q4/5

True or false?

An inscribed triangle can have a vertex inside the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed triangle is equilateral • High School — Geometry • Geometry: Circles • HSG-C.A.2

Points A, B, C, D, E, F are marked around a circle with center O by stepping the radius. Prove that triangle ACE is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q1/5

A circle has center O and diameter AD. An arc centered at D with radius DO meets the circle at B and C. Prove that triangle ABC is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q2/5

Explain why stepping a compass set to the radius around a circle returns exactly to the starting point after six steps.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q3/5

In the stepped construction, points A, B, C, D, E, F lie on a circle with center O, each 60° from the next. Prove that ∠CAE = 60°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q4/5

Ana inscribes a triangle by marking points every 120° around the center with a protractor. Explain why her triangle is equilateral, and why her method is not a compass-and-straightedge construction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 34 • FORWARD 1 • Q5/5

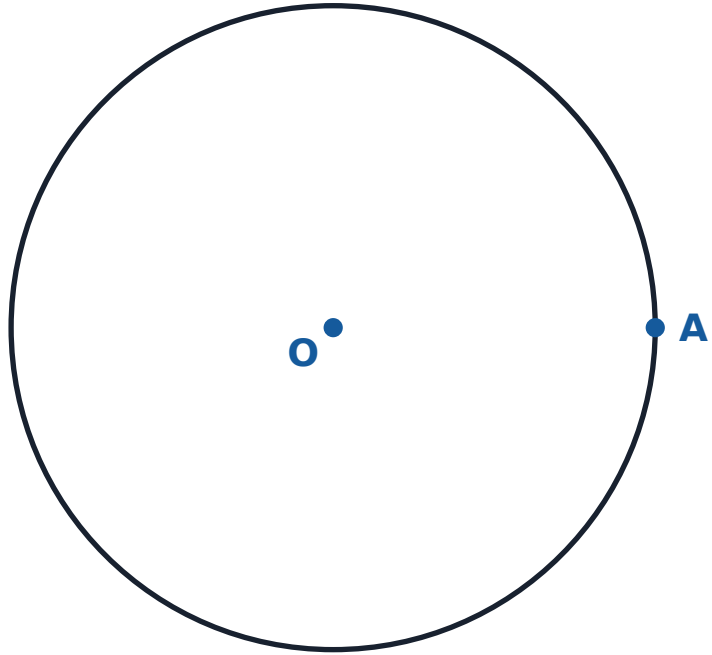
SET 34 • HSG-CO.D.13

FORWARD 2

Construct a regular dodecagon by bisecting arcs • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular 12-sided polygon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q1/5

Each side of the inscribed regular hexagon subtends 60° at the center. What does each side of the dodecagon subtend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q2/5

True or false?

Bisecting the arcs of an inscribed regular hexagon gives the vertices of a regular nonagon.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q3/5

Starting from an inscribed square, bisecting arcs gives a regular

A. pentagon

B. hexagon

C. decagon

D. octagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q4/5

How many sides does the polygon have after bisecting the arcs of an inscribed equilateral triangle twice?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a square inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct a perpendicular bisector	**HSG-CO.D.12**
BACKWARD 2	Diagonals of a square	**HSG-CO.C.11**
BACKWARD 3	Diameters and the center of a circle	**7.G.B.4**
FORWARD 1	Justify that the inscribed quadrilateral is a square	**HSG-CO.C.11**
FORWARD 2	Construct a regular octagon inscribed in a circle	**HSG-CO.D.13**

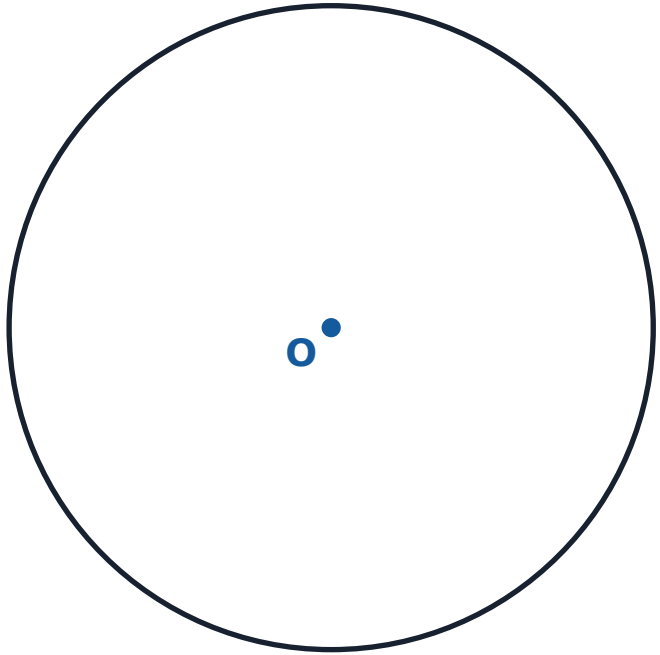
SET 35 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle.



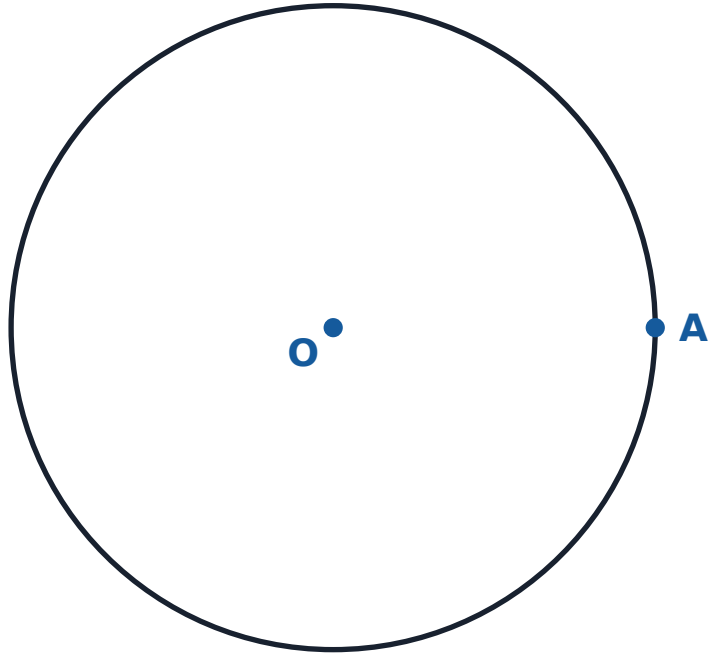
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one vertex at A.



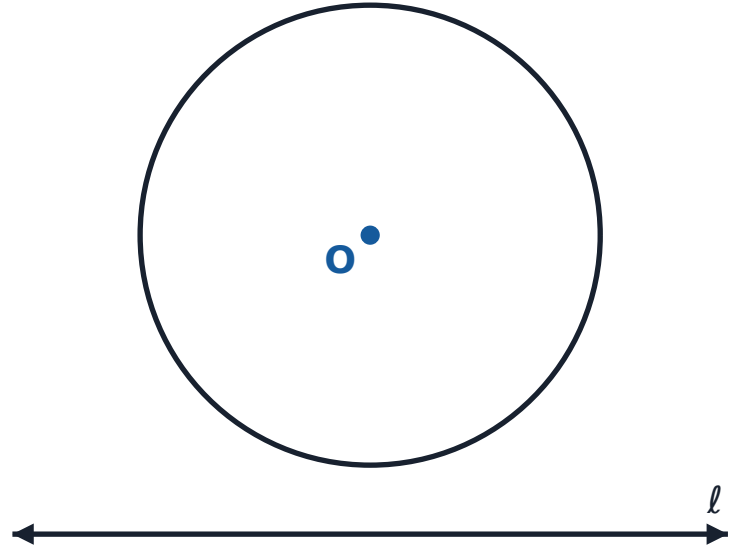
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle with one side parallel to the given line ℓ.



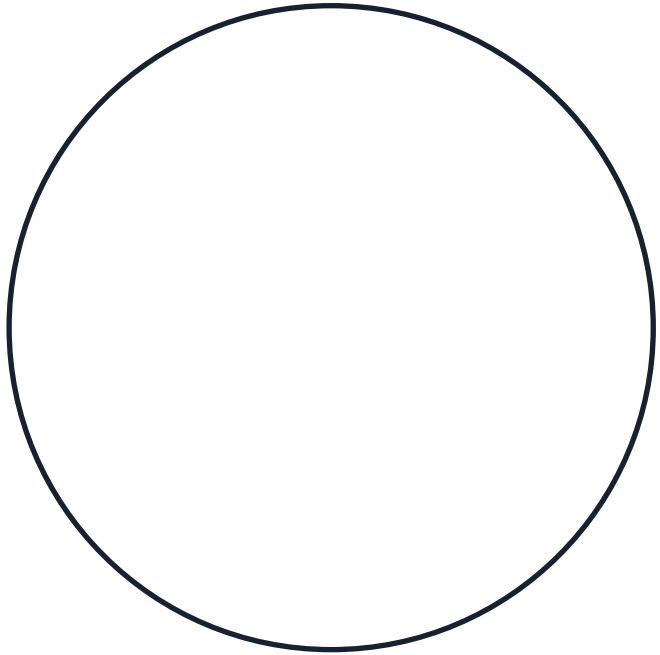
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a square inscribed in it.



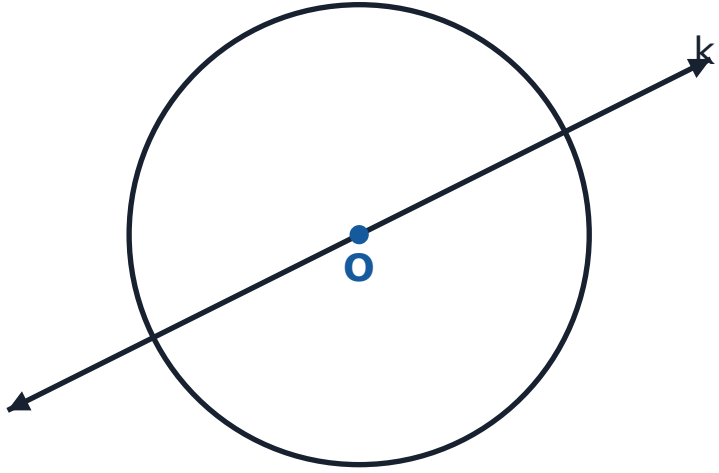
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

Construct a square inscribed in the circle so that one of its diagonals lies along line k.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • MAIN • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 1

Construct a perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.D.12

Which construction turns one diameter into two perpendicular diameters?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of a diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q2/5

Two perpendicular diameters cut the circle into

A. six 60° arcs

B. four 45° arcs

C. four 90° arcs

D. two 90° arcs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q3/5

A diameter is 10 cm. Where does its perpendicular bisector cross it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q4/5

True or false?

A perpendicular bisector can be drawn by measuring with a ruler only.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.12 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 2

Diagonals of a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

True or false?

The diagonals of a square are congruent and perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q1/5

A quadrilateral's diagonals are congruent, perpendicular, and bisect each other. What is its most specific name?

A. rhombus that is not a square

B. square

C. trapezoid

D. kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q2/5

A square has diagonal 8. Find its side.

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q3/5

True or false?

A rectangle always has perpendicular diagonals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q4/5

The diagonals of a square meet at O. Name the angle between them.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSG-CO.D.13

BACKWARD 3

Diameters and the center of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has radius 3. How long is any diameter?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q1/5

True or false?

Every diameter passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q2/5

Where do two different diameters of a circle meet?

A. At the midpoint of a radius

B. On the circle

C. At the center

D. Nowhere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q3/5

What is the longest chord of a circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q4/5

True or false?

All diameters of a circle have different lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed quadrilateral is a square • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

Diameters AC and BD of a circle with center O are perpendicular. Prove that ABCD is a square.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q1/5

A square is inscribed in a circle of radius 5. Find its side length and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q2/5

True or false?

The side of a square inscribed in a circle of radius r is 2r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q3/5

The area of a square inscribed in a circle of radius r is

A. $2r^2$

B. πr^2

C. $4r^2$

D. r^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q4/5

What fraction of the circle's area does the inscribed square cover? Give the exact value.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 35 • FORWARD 1 • Q5/5

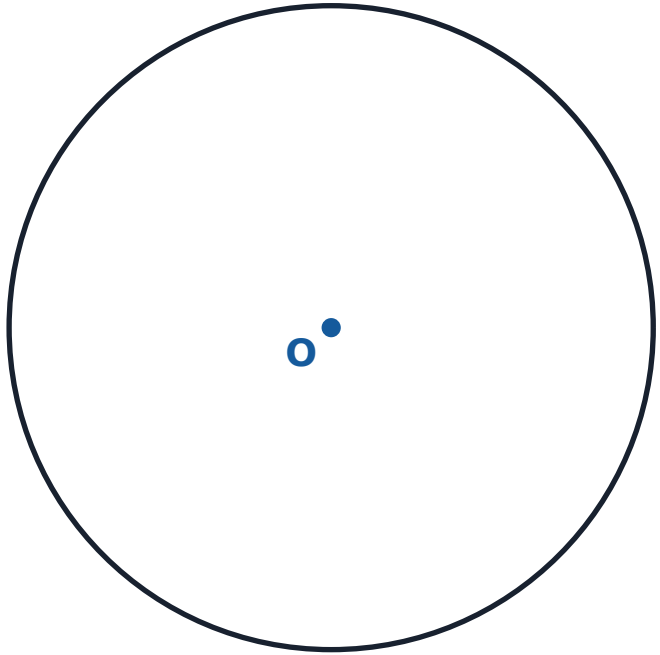
SET 35 • HSG-CO.D.13

FORWARD 2

Construct a regular octagon inscribed in a circle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular octagon inscribed in the circle.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q1/5

What is the central angle for each side of a regular octagon?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q2/5

True or false?

A regular octagon has interior angles of 120°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q3/5

Bisecting the arcs of an inscribed regular octagon gives a regular

A. hexagon

B. 16-gon

C. 12-gon

D. square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q4/5

A regular octagon is constructed in a circle by bisecting the four 90° central angles of an inscribed square. Explain why the eight points are equally spaced.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSG-CO.D.13 • CONGRUENCE: CONSTRUCTIONS

Construct a regular hexagon inscribed in a circle

MAIN	Geometry main question	**HSG-CO.D.13**
BACKWARD 1	Construct an inscribed equilateral triangle	**HSG-CO.D.13**
BACKWARD 2	An equilateral triangle has equal sides and 60° angles	**4.G.A.2**
BACKWARD 3	Circle facts: radius and chord	**7.G.B.4**
FORWARD 1	Justify that the inscribed hexagon is regular	**HSG-C.A.2**
FORWARD 2	Use inscribed hexagons to estimate π	**HSG-GMD.A.1**

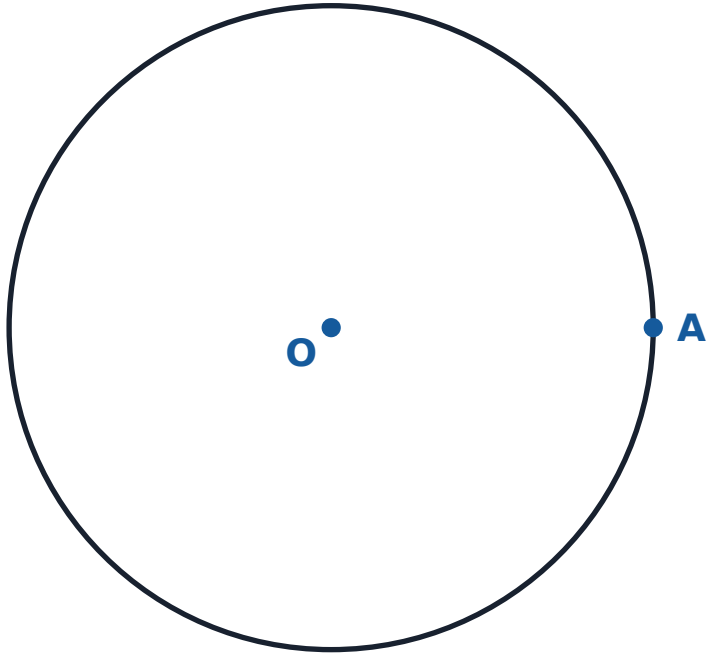
SET 36 • HSG-CO.D.13

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle, with one vertex at A.



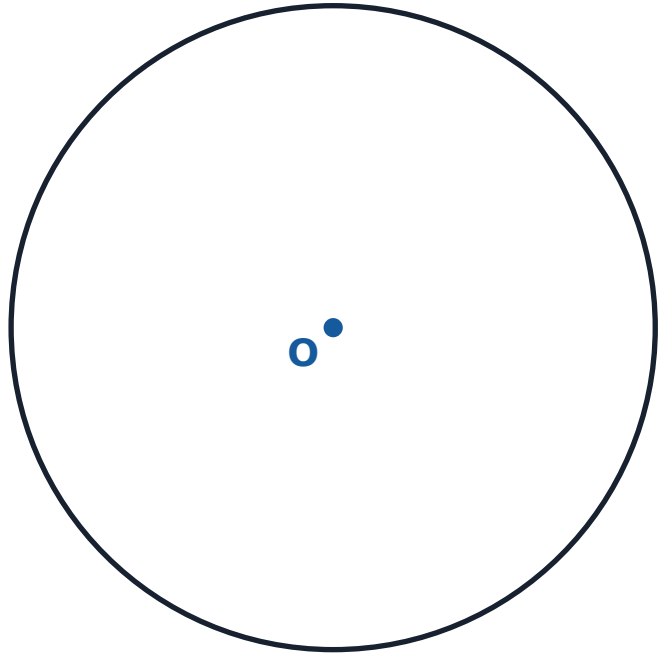
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q1/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle (choose any starting point).



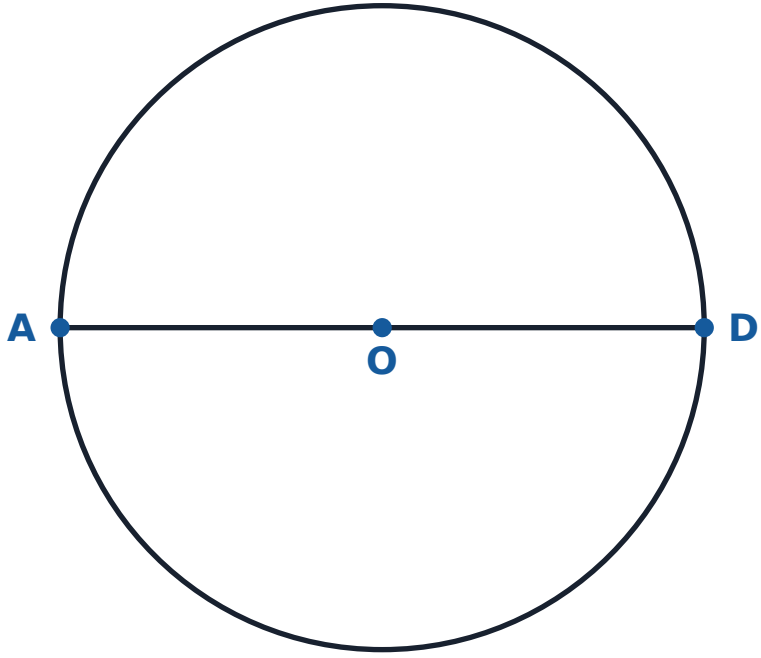
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q2/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle using diameter AD and arcs centered at A and D with radius equal to the circle's radius.



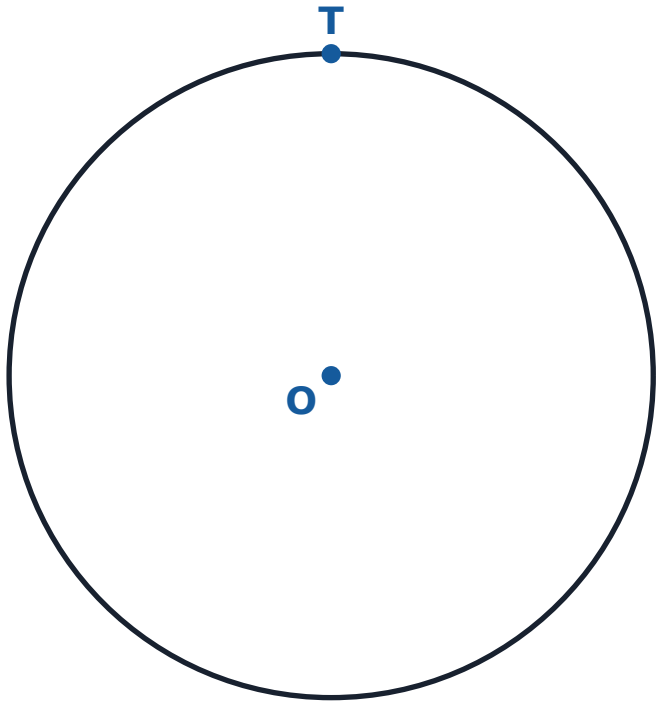
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q3/5

Compass and straightedge only; leave all arcs.

Construct a regular hexagon inscribed in the circle with one vertex at the top point T.



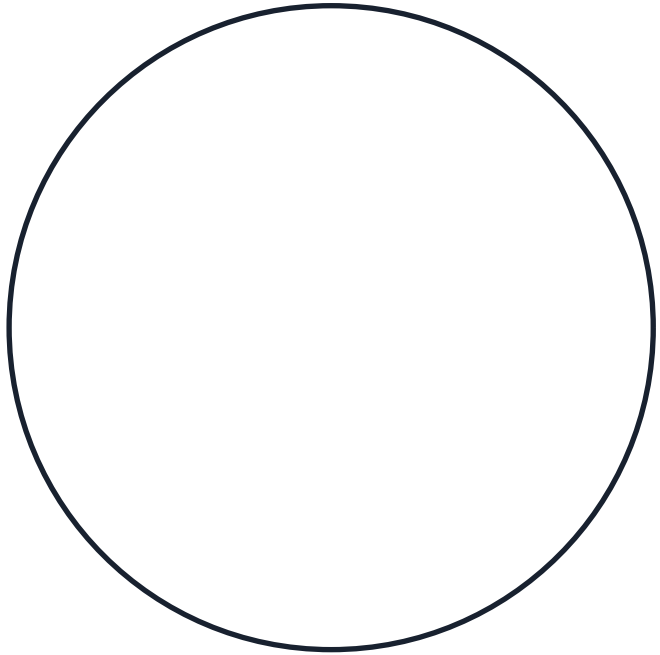
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q4/5

Compass and straightedge only; leave all arcs.

The center of this circle is not marked. Construct a regular hexagon inscribed in it.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • MAIN • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 1

Construct an inscribed equilateral triangle • High School — Geometry • Geometry: Congruence • HSG-CO.D.13

How are the inscribed equilateral triangle and the inscribed regular hexagon related?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q1/5

True or false?

Both constructions start by stepping the radius around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q2/5

How many points does stepping the radius mark?

A. 12

B. 6

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q3/5

An equilateral triangle is inscribed. How many more points must be added to make a regular hexagon?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q4/5

True or false?

The inscribed equilateral triangle and hexagon have the same side length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.D.13 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 2

An equilateral triangle has equal sides and 60° angles • Grade 4 • Geometry • 4.G.A.2

True or false?

Each angle of an equilateral triangle measures 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q1/5

A triangle has three sides equal to the radius of a circle. What kind of triangle is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q2/5

Six equilateral triangles meeting at a point fill an angle of

A. 300°

B. 360°

C. 60°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q3/5

A regular hexagon has side 5. How long are the segments from its center to its vertices?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon splits into four equilateral triangles from its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSG-CO.D.13

BACKWARD 3

Circle facts: radius and chord • Grade 7 • Geometry • 7.G.B.4

A chord equals the radius. What central angle does it subtend?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q1/5

True or false?

A chord equal to the radius forms an equilateral triangle with the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q2/5

A circle has radius 7. A regular inscribed hexagon has perimeter

A. 21

B. 14

C. 42

D. 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q3/5

A circle has circumference 31.4 (π ≈ 3.14). Find its radius.

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q4/5

True or false?

Every chord of a circle is as long as the radius.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 1

Justify that the inscribed hexagon is regular • High School — Geometry • Geometry: Circles • HSG-C.A.2

Six points are marked around a circle with center O by stepping the radius. Prove that the hexagon they form is regular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q1/5

A compass set to the radius of a circle is stepped around the circle from a starting point. Why does the sixth step land exactly back on the starting point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q2/5

True or false?

Each interior angle of a regular hexagon is 60°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q3/5

A regular hexagon inscribed in a circle of radius 4 has perimeter

A. 12

B. 48

C. 24

D. 8π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q4/5

Find the area of a regular hexagon inscribed in a circle of radius 2.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSG-CO.D.13

FORWARD 2

Use inscribed hexagons to estimate π • High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A regular hexagon is inscribed in a circle of radius r. Compare its perimeter with the circumference. What estimate of π does this give?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q1/5

A regular 12-gon inscribed in a circle of radius 1 has perimeter about 6.212. What estimate of π does it give, to the nearest hundredth?

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q2/5

True or false?

Doubling the number of sides of an inscribed regular polygon makes its perimeter longer than the circumference.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q3/5

Which polygon's perimeter is closest to the circumference of its circle?

A. Inscribed regular 96-gon

B. Inscribed regular hexagon

C. Inscribed square

D. Inscribed triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q4/5

Explain why an inscribed polygon's perimeter is always less than the circumference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Geometry • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 36 • FORWARD 2 • Q5/5

Verify that a dilation maps a line to a parallel line (or to itself through the center)

MAIN	Geometry main question	**HSG-SRT.A.1.a**
BACKWARD 1	Dilate points with coordinates	**8.G.A.3**
BACKWARD 2	Compare slopes to test whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Write the equation of a line through two points	**8.F.B.4**
FORWARD 1	Dilate lines about a center other than the origin	**HSG-SRT.A.1.a**
FORWARD 2	Prove algebraically that a dilation maps a line to a parallel line	**HSG-SRT.A.1.a**

SET 37 • HSG-SRT.A.1.a

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Line ℓ is $y = 2x + 1$ (it does not pass through the origin). Dilate it about the origin with scale factor 3: find the images of (0, 1) and (1, 3), write an equation of the image line, and state what this example shows about dilations and lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q1/5

Line m is $y = -x$, which passes through the origin. Dilate it about the origin with scale factor 2: find the images of $(1, -1)$ and $(3, -3)$. What is the image line, and what does this example suggest about lines through the center?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q2/5

The horizontal line y = 4 is dilated about the center C(0, 2) with scale factor ½. Find the images of (0, 4) and (6, 4), write an equation of the image line, and compare it with the original.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q3/5

Line n is y = −½x + 4. Dilate it about the center C(2, 1) with scale factor 3: find the images of (0, 4) and (4, 2), write an equation of the image line, and compare it with n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q4/5

Line p is y = 2x, which passes through C(1, 2). Dilate it about C with scale factor 4: find the images of (0, 0) and (2, 4) and the image line. What do you observe?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • MAIN • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 1

Dilate points with coordinates • Grade 8 • Geometry • 8.G.A.3

Dilate $(2, -3)$ about the origin with scale factor 4.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q1/5

Dilate (6, 10) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q2/5

A dilation about the origin sends (3, 1) to (9, 3). What is the scale factor?

A. 6

B. 9

C. 3

D. ⅓

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q3/5

True or false?

A dilation about the origin with scale factor 1 leaves every point where it is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q4/5

True or false?

A dilation with scale factor 2 moves the center to a new point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 2

Compare slopes to test whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (0, 3) and (3, 9).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q1/5

True or false?

$y = 2x + 1$ and $y = 2x + 3$ are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q2/5

Which line is parallel to $y = -3x + 2$?

A. $y = \frac{1}{3}x$

B. $y = -3$

C. $y = -3x - 5$

D. $y = 3x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q3/5

Are y = 5 and y = 2 parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q4/5

True or false?

Lines with slopes 2 and −2 are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSG-SRT.A.1.a

BACKWARD 3

Write the equation of a line through two points • Grade 8 • Functions • 8.F.B.4

Write an equation of the line through (0, 3) and (3, 9).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q1/5

Write an equation of the line through (2, −2) and (6, −6).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q2/5

Which point is on y = 2x + 3?

A. (5, 1)

B. (2, 3)

C. (0, 2)

D. (1, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q3/5

True or false?

The line through (0, 0) and (4, 8) is $y = 2x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q4/5

True or false?

The line through (1, 1) and (2, 4) has slope 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 8 • Functions • 8.F.B.4 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 1

Dilate lines about a center other than the origin • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Dilate the line $y = x + 2$ about the center $(2, 0)$ with scale factor 2. Find the images of $(0, 2)$ and $(2, 4)$ and the image line.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q1/5

Does the line $y = x - 2$ pass through the center $(2, 0)$? What happens to it under any dilation about $(2, 0)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q2/5

True or false?

Under a dilation about (1, 1) with scale factor 3, the point (2, 1) goes to (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q3/5

A dilation about (0, 2) with scale factor ½ maps (4, 6) to

A. (4, 4)

B. (8, 10)

C. (2, 3)

D. (2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q4/5

Line ℓ: y = 3x. A dilation about (1, 0) with scale factor 2 maps ℓ to ℓ'. Find ℓ'.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSG-SRT.A.1.a

FORWARD 2

Prove algebraically that a dilation maps a line to a parallel line • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a

Prove that the dilation about the origin with scale factor k ($k > 0, k \neq 1$) maps the line $y = mx + b$ to the line $y = mx + kb$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q1/5

From the general result, when does the image of $y = mx + b$ coincide with the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q2/5

True or false?

The dilation $(x, y) \to (kx, ky)$ maps the vertical line $x = c$ to the line $x = c + k$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q3/5

Under the dilation about the origin with scale factor 4, the line $y = -2x + 5$ maps to

A. $y = -8x + 5$

B. $y = -2x + 20$

C. $y = -2x + 5$

D. $y = -8x + 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q4/5

Explain why measuring a few dilated lines is not a proof of the property.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.a • Set 37 • FORWARD 2 • Q5/5

Verify that a dilation multiplies lengths by the scale factor

MAIN	Geometry main question	**HSG-SRT.A.1.b**
BACKWARD 1	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 2	Scale factor in scale drawings	**7.G.A.1**
BACKWARD 3	Dilate points about the origin	**8.G.A.3**
FORWARD 1	Find the center or scale factor from a segment and its image	**HSG-SRT.A.1.b**
FORWARD 2	Perimeter and area of dilated figures	**HSG-SRT.A.2**

SET 38 • HSG-SRT.A.1.b

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

Segment AB has A(1, 2) and B(4, 6). Dilate it about the origin with scale factor 3. Find A′B′ and AB, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q1/5

Segment PQ has P(−4, 2) and Q(8, −6). Dilate it about the origin with scale factor ½. Find PQ and P′Q′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q2/5

Segment CD has C(2, 1) and D(5, 5). Dilate it about the center (2, 1) with scale factor 2. Find C′D′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q3/5

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Dilate it about (1, 1) with scale factor 2. Find all three image side lengths, compare each with the original, and say what the comparison supports.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q4/5

Segment EF has E(−3, −1) and F(3, 7). Dilate it about the origin with scale factor 0.4. Find EF and E′F′, and compare.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • MAIN • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 1

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (3, 6) and (12, 18).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 1)$ and $(4, -3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

A. $\sqrt{17}$

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q3/5

True or false?

The distance between (1, 2) and (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance between (0, 0) and (3, 3) is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 2

Scale factor in scale drawings • Grade 7 • Geometry • 7.G.A.1

A drawing uses scale factor 4. A 3 cm segment in the original is how long in the drawing?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q1/5

A 15 m wall is drawn 5 cm long. What scale factor (drawing ÷ actual, in the same units) was used?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q2/5

True or false?

A scale factor less than 1 makes a figure smaller.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q3/5

A rectangle 4 by 6 is enlarged by a scale factor of 1.5. Its new dimensions are

A. 5.5 by 7.5

B. 4 by 9

C. 8 by 12

D. 6 by 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q4/5

True or false?

Enlarging by a scale factor of 2 doubles the area.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSG-SRT.A.1.b

BACKWARD 3

Dilate points about the origin • Grade 8 • Geometry • 8.G.A.3

Dilate (4, 6) about the origin with scale factor 3.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q1/5

Dilate (−4, 2) about the origin with scale factor ½.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q2/5

Dilation about the origin sends (2, 5) to (5, 12.5). Scale factor?

A. 2.5

B. 3

C. 0.4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q3/5

True or false?

Dilating about the origin multiplies both coordinates by the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q4/5

True or false?

Dilating (3, 0) by 2 about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 1

Find the center or scale factor from a segment and its image • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b

A dilation about the origin maps a segment of length 6 to a segment of length 15. What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q1/5

A dilation maps A(1, 1) to A′(3, 3) and B(2, 1) to B′(5, 3). Find the scale factor and the center.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q2/5

True or false?

Under a dilation with scale factor $k \neq 1$, the line through any point P other than the center and its image P′ passes through the center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q3/5

A dilation maps P(4, 0) to P′(1, 0) and its center is the origin. Scale factor?

A. 4

B. −3

C. ¼

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q4/5

A segment of length 9 is dilated by an unknown factor k, and the image is 3 units shorter than the original. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1.b • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSG-SRT.A.1.b

FORWARD 2

Perimeter and area of dilated figures • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

A triangle with perimeter 18 and area 12 is dilated with scale factor 3. Find the new perimeter and area.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q1/5

A square of area 50 is dilated so its area becomes 8. Find the scale factor.

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q2/5

True or false?

If lengths are multiplied by k, areas are multiplied by 2k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q3/5

A figure is dilated with scale factor ½. Its area is multiplied by

A. ½

B. 4

C. 2

D. ¼

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q4/5

Explain why a dilation with factor k multiplies the perimeter by k.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 38 • FORWARD 2 • Q5/5

Decide whether two figures are similar using similarity transformations

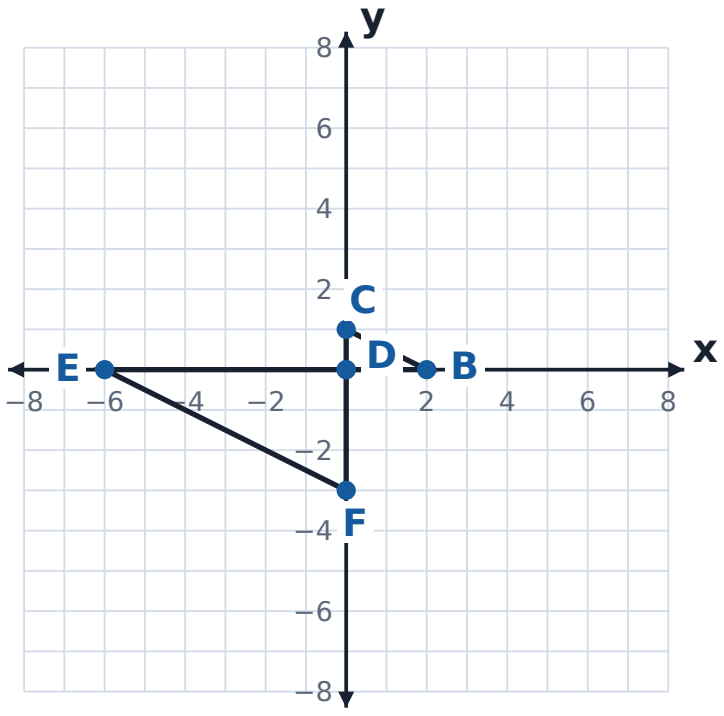
MAIN	Geometry main question	**HSG-SRT.A.2**
BACKWARD 1	Similar figures as a sequence of transformations including a dilati…	**8.G.A.4**
BACKWARD 2	Specify a sequence of rigid motions	**HSG-CO.A.5**
BACKWARD 3	Dilate figures about the origin	**8.G.A.3**
FORWARD 1	Find the scale factor and correspondence when the image is turne	**HSG-SRT.A.2**
FORWARD 2	Prove that all circles are similar	**HSG-C.A.1**

SET 39 • HSG-SRT.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Is triangle ABC similar to triangle DEF? If so, describe a similarity transformation that maps ABC onto DEF; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 39 • MAIN • Q1/5

Is rectangle R (4 by 6) similar to rectangle S (6 by 8)? Explain using similarity transformations.

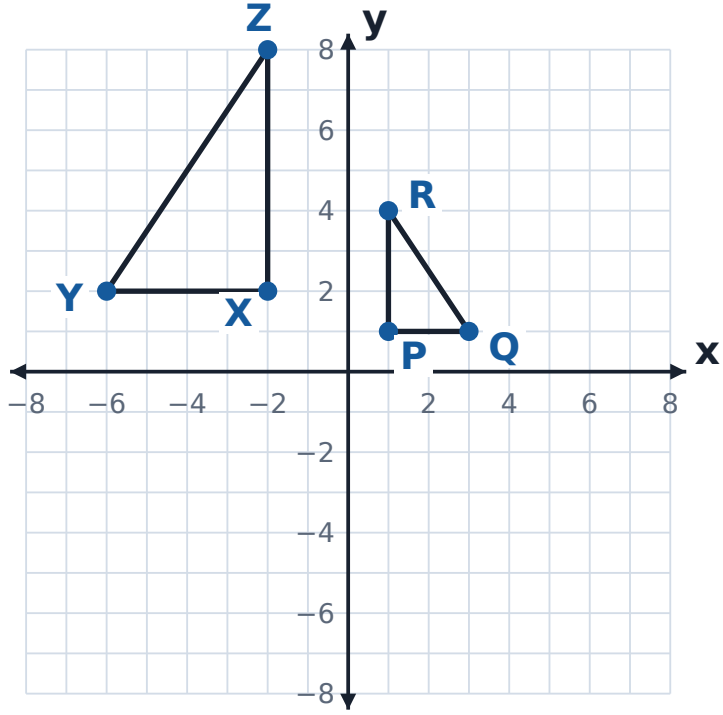
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

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Is triangle PQR similar to triangle XYZ? If so, describe a similarity transformation; if not, explain.



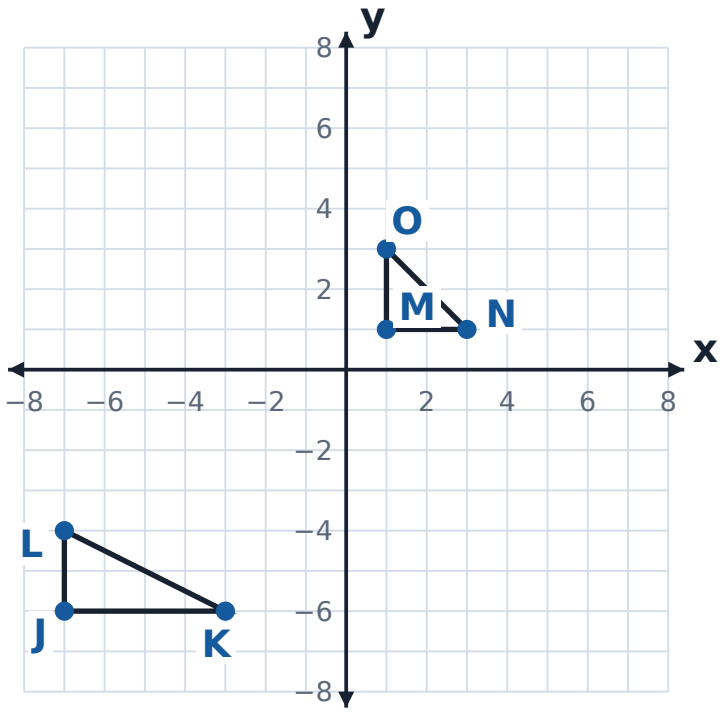
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

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Is triangle JKL similar to triangle MNO? If so, describe a similarity transformation; if not, explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

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